Design Narrative

Replace Industrial Waste Lift Stations 6n & 6S, B62501

Project Number: WWYK240176
Work Order: 15311657

Tinker Air Force Base, Oklahoma

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Prepared By:

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CONTENTS

APPENDICES

1. Overview

This project replaces the existing Industrial Waste Lift Stations 6N and 6S at Building 62501 with a new system that meets current standards and provides reliable service. Work includes removing the existing stations, installing new lift station equipment and valve vaults, and connecting them to the existing industrial waste system. Mechanical upgrades include a new wet well, new submersible pumps, containment piping, and improved screening to manage future flows, and maintenance needs. Site grading and drainage improvements will support the new installation. Electrical upgrades will include new controls, power improvements, and integration with the base monitoring system. Construction will be sequenced to keep one lift station in service at all times and minimize disruptions.

2. Applicable Codes and Standards

2.1 General

- ICD/ICS 705 — Technical Specifications for Construction and Management of Sensitive Compartmented Information Facilities
- IBC — International Building Code (as amended by UFC 1-200-01)
- IEBC — International Existing Building Code
- NFPA 101 — Life Safety Code
- TAFB Facility Standard
- UFC 1-200-01 — DoD Building Code
- UFC 4-010-01 — DoD Minimum Antiterrorism Standards for Buildings
- UFC 4-010-05 — Sensitive Compartmented Information Facilities Planning, Design, and Construction

2.2 Civil

- FAA Advisory Circular 150/5340-1L
- BASH Manual
- UFC 3-201-01 — Civil Engineering
- UFC 3-240-01 — Wastewater Collection and Treatment
- UFC 3-230-01 — Water Storage, Distribution, and Transmission
- UFC 3-250-01 — Pavement Design for Roads and Parking Areas
- UFC 4-022-02 — Security Fences and Gates

2.3 Architectural

- ABA — Architectural Barriers Act Standards
- UFAS — Uniform Federal Accessibility Standards
- UFC 3-101-01 — Architecture
- UFC 3-110-03 — Roofing
- UFC 3-120-10 — Interior Design
- UFC 3-190-06 — Protective Coatings and Paints
- UFC 3-201-02 — Landscape Architecture

2.4 Structural

- ACI 318-14 — Building Code Requirements for Structural Concrete and Commentary
- AISC 360-10 — Steel Construction Manual, 14th Ed.
- ASCE 7-10 — Minimum Design Loads for Building and other Structures
- AISI S100-12 — North American Specification for the Design of Cold-formed Steel Structural Members
- UFC 3-301-01 — Structural Engineering
- UFC 3-310-01 — Seismic Design of Buildings

2.5 Mechanical / Plumbing

- ANSI/ASHRAE/ ACCA Standard 183 — Peak Cooling and Heating Load Calculations in Buildings Except Low-Rise Residential Buildings
- ASHRAE 55 — Thermal Environmental Conditions for Human Occupancy
- ASHRAE 62.1 — Ventilation for Acceptable Indoor Air Quality
- ASHRAE 90.1 — Energy Standard for Buildings Except Low-Rise Residential Buildings
- IECC — International Energy Conservation Code
- IMC — International Mechanical Code (as amended by UFC 3-410-01)
- IPC — International Plumbing Code (as amended by UFC 3-420-01)
- TAFB Mechanical Standard
- UFC 3-401-01 — Mechanical Engineering
- UFC 3-410-01 — Heating, Ventilating, and Air Conditioning Systems
- UFC 3-410-02 — Direct Digital Control for HVAC and Other Building Control Systems
- UFC 3-410-04 — Industrial Ventilation
- UFC 3-420-01 — Plumbing Systems
- UFC 3-430-01FA — Heating and Cooling Distribution Systems
- UFC 3-450-01 — Noise and Vibration Control

2.6 Electrical

- NFPA 70 — National Electrical Code
- NFPA 780 — Standard for Installation of Lightning Protection Systems
- TAFB Electrical Standard
- UFC 3-501-01 — Electrical Engineering
- UFC 3-520-01 — Interior Electrical Systems
- UFC 3-530-01 — Interior and Exterior Lighting Systems and Controls
- UFC 3-575-01 — Lightning and Static Electricity Protection Systems
- AFMAN 32-1065 — Grounding and Electrical Systems

3. Civil Design Narrative

The civil requirements for the project include (but are not limited to):

- Demolish and remove the existing lift station structures.
- Prepare the site to accommodate the installation of new lift stations and valve vaults.
- Coordinate with existing utilities to avoid conflicts during construction.
- Provide stable, properly graded surfaces around the new structures.
- Improve drainage and maintenance access.
- Minimize service interruptions during tie-in activities.

3.1 Existing Conditions

The project area includes two existing industrial waste lift stations, 6N to the north and 6S to the south, both located within a fenced area. Lift Station 6N currently receives the majority of the industrial waste flow, while 6S only receives a smaller tributary amount. The site is mostly covered with grass and enclosed by fencing, with gravel access provided to 6N. Drainage generally flows away from the lift stations, though some localized grading improvements are needed to prevent ponding.

3.2 Design Approach

The civil design focuses on rerouting the flow from Lift Station 6S into Lift Station 6N, then removing the old lift stations and installing a new lift station at the former 6S location. The flow reroute will be completed first to maintain continuous system operation. After rerouting, 6S will be demolished, and the existing pit will be reused for the new lift station. Once the new lift station is operational, 6N will be demolished.

3.3 Grading and Drainage

Grading will be designed to provide positive drainage away from the new lift station areas, avoiding low spots and ponding. Concrete pads will be constructed around the new lift stations and valve vaults to provide stable bases and improved maintenance access, while gravel walkways will be added to further support operations.

3.4 Environmental Management (EM) Survey

During the demolition of the industrial waste station, it is presumed that asbestos-containing materials (ACM) and/or lead-based paint (LBP) will be present. The contractor is required to test for these materials once they are exposed. If the contractor encounters any suspect environmental conditions at any time, they must immediately notify the Contracting Officer (CO) and the Contracting Officer's Representative (COR).

An EM survey indicates there are no asbestos containing materials (ACM), heavy metals (HM or lead based paint (LBP) concerns for area(s) where work is to be performed. If at any time the Contractor encounters suspect environmental conditions, they shall notify the CO and the COR.

(a) Any dewatering needs to be pumped and hauled to the IWTP

(b) Any excavation needs to be stored in a lined container, tested, and disposed of in accordance with the sampling results. (Ref. Tinker Hazardous Waste Management Plan).

If at any time the Contractor encounters suspect environmental conditions not identified above, they shall notify the CO and the COR.

3.5 Design Considerations and Assumptions

Utility records and topographic data will be verified in the field before final design to confirm accuracy. Existing soils are assumed to have adequate bearing capacity for proposed structures. Dewatering may be required during excavation depending on groundwater conditions. Construction will be sequenced so that at least one lift station remains operational at all times. All work will comply with UFC 3-201-01 (Civil Engineering), UFC 3-230-01 (Pumping Facilities), UFC 3-240-01 (Wastewater Collection and Treatment), EPA regulations, and Tinker AFB design standards.

4. Structural Design Narrative

4.1 General Design Parameters and Assumptions

4.1.1 Concrete and Reinforcing Steel

- Structural Cast-In-Place Concrete, Unless Otherwise Indicated: 4000 psi.
- Concrete for Paving, Sidewalks, slab-on-grade, and Other Flatwork: 4000 psi
- Cement: Type II or I/II for cast-in-place concrete.
- Reinforcing Steel (Deformed Bars): ASTM A615, Grade 60.
- Reinforcing Steel to Be Welded (Deformed Bars): ASTM A706, Grade 60.
- Anchor Bolts: ASTM F1554, Gr 36, Galvanized.

4.1.2 Reinforced Concrete Design

Design of Slab-On-Ground: Slabs-on-ground shall be designed in accordance with ACI 360R-10 and ACI 318-14.

4.1.3 Structural Steel

- W and WT Shapes: ASTM A992 or ASTM A572, Grade 50, with special requirements per AISC Technical Bulletin #3.
- Other Rolled Sections and Plates: ASTM A36 or ASTM A572, Grade 50.
- Square or Rectangular Tubes (HSS): ASTM A500, Grade B.
- High Strength Bolts: ASTM A325, with compressible washer type direct tension indicators or twist of type tension control bolts.
- Anchor Bolts: ASTM F593, Type 304.

4.1.4 Steel Design

Structural steel design will be performed in accordance with the requirements of the American Institute of *Manual Steel Construction (14th Edition)*, AISC 360-10.

4.1.5 Structural Loads

4.1.5.1 Dead Loads

Dead loads will be considered resulting from fixed construction, materials, and equipment. Dead loads will be determined and applied in accordance with ASCE 7-10, and other pertinent standards and vendor-supplied information.

4.1.5.2 Live Loads

Live loads are those loads produced by the use and occupancy of the building or structure, and do not include loads such as wind/seismic loads, rain loads, flood loads, or dead loads. Minimum uniformly distributed live loads will be as required by ASCE 7-10.

4.1.5.3 Wind Loads

General: Wind loads for structural design of main wind force resisting systems and components and cladding shall be determined and applied in accordance with ASCE 7-10, Directional procedure.

Wind Speed: The minimum 3-second gust design wind speed shall be determined from Figure 26.5 of ASCE 7-10. A basic wind speed of 120 mph will be appropriate for the project location.

Exposure Category: Building exposure classification C, as defined by ASCE 7-10, shall be used for this building.

Risk Category: Building risk category IV, as defined by ASCE 7-10, shall be used for this building.

4.1.5.4 Load Combinations

Load combinations using strength design and service-level combinations shall be as defined in ASCE 7-10.

4.1.5.5 Seismic and Snow Loads

Seismic loads shall be determined in accordance with IBC 2015. The structure will be designed as Seismic Design Category of “C” based on the seismic parameters defined in IBC 2015.

Snow loads shall be determined in accordance with IBC 2015. The ground snow load shall be 10 pounds per square foot.

4.2 Structural Demolition- N/A

4.3 Structural Construction- N/A

5. Mechanical / Plumbing Design Narrative

The mechanical requirements for the project include (but are not limited to) Industrial Waste Lift Station (IWLS), Industrial Waste Valve Vault (IWVV), and Meter Vault (MV):

- Installation shall be designed and constructed IAW UFC 3-420-01 Plumbing Systems, UFC 3-240-01 Domestic Wastewater Treatment, and the TAFB Mechanical Standard.
- Provide sump of approximately 5,000-gallon capacity (10' diameter X 22' depth) with three submersible pumps for 3" spheres, DOR to confirm 100 GPM, 10 HP pumps with N+1 system redundancy. Pump flow rate to ensure 2 ft/sec in outlet piping. Provide inlet strainer basket compatible with pump flow rate and strainer mesh size for current influent paint chip size.
- Provide flow meter on pump discharge piping in new meter vault on each IW pipe that routes to the IWTP.
- Provide piping of existing size and configuration to include containment to and from all inlet sources to 6N & 6S to new IWLS. Connect to existing IW piping from new IWVV with containment piping. Connect existing outlet piping with containment to the IWVV. Configure piping to use either pipe to the IW Treatment Plant.

5.1 General Design Parameters and Assumptions

5.1.1 Elevation, Latitude/Longitude, Climate Zone

Elevation, latitude/longitude, and climate zone were taken from International Energy Conservation Code 2015. Weather data below was taken from UFC 3-400-02 and 2021 ASHRAE Fundamentals Chapter 14.

- Elevation: 1,260 ft.
- Latitude: 35.42 N
- Longitude: 97.38 W
- Climate Zone: 3
- Summer: 100°F DB/74°F WB
- Winter: 13.9°F

5.1.2 General Indoor Temperature/Relative Humidity Parameters: N/A

5.2 Building HVAC Loads: N/A

5.3 Indoor Air Quality: N/A

5.4 Validation of Existing HVAC Systems: N/A

5.5 Life Cycle Cost Analysis (LCCA)

Submersible pumps are the most applicable type of pumps for this service.

Based on the information above, an LCCA was not performed for this project.

5.6 Demolition

Removal of the existing pumps and piping within each sump 6N & 6S.

5.7 Mechanical Construction

5.7.1 Industrial Waste Lift Station

Install three (3) submersible pumps in 10’ diameter by 22’ deep FRP (Fiberglass Reinforced Plastic) sump with rail system. Pump system to provide N+1 criteria to maintain two working pumps. Provide piping to route through either existing 6” IW to the IWTP with check valves, shut-off valves, and flow metering.

5.7.2 Vibration Control – N/A

5.7.3 Energy Conservation

The following energy conservation measures are being used on this project.

- Pump motor VFD with integral bypass for soft start purposes.

5.7.4 Controls

Local control will be provided on all new submersible pumps. Pumps will be controlled by packaged pump effluent level control system. The direct digital controls will be tied into the base side COINe (Community of Interest Network Enclave) and Honeywell EBI DDC system located in B2136 to allow monitoring and troubleshooting remotely as well as at the facility. New graphics will be provided for all new pumps and flow meters.

5.7.5 System Commissioning, Testing and Balancing

Provide commissioning of work designed/constructed under this project. Also provide to a firm certified by the Associated Air Balance Council (AABC) or National Environmental Balancing Bureau (NEBB) to perform the testing, adjusting and balancing procedures of air side and water side systems per the requirements of the AABC or NEBB Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems.

6. Electrical Design Narrative

Electrical design shall confirm the following:

- Number and size of pumps (gpm, horsepower, and electrical demand in kilowatts)
- Location(s) or the Equipment Rack and associated electrical box(s) above grade or in the vault(s).
- Verify that the existing emergency power system with automatic transfer switch systems for alternative power generation and distribution in an emergency can support the new operations.
- Provide a motor starting analysis and maximum short circuit calculations. Lists maximum short circuit amperage on the drawings and specifies the method of calculation in accordance with Tinker AFB Standards.
- The design engineer shall specify a power system study and ARC fault labeling according to Tinker AFB Standards.
- Preform electrical cable testing on the existing cables' performance, insulation conductivity, and safety under the various electrical conditions for this lift station upgrade.

6.1 General Design Parameters and Assumptions

The existing electric service for Industrial Waste Lift Station 6N & 6S (Facility B62502) is fed from Panel MDP 2119-PNL-6, which is a 480/277 Volt, 3 Phase, 100 Amp Service (Circuit 1, 3, 5) in (Facility 2119). This panel is on an Emergency Generator.

All enclosures shall be stainless steel 316 (or equal), rated NEMA 4X with level door closures and aluminum dead-front covers. The pump control enclosure shall be of the double door type to locate all 480V equipment on one side and all 120V equipment on the other side. Include a Wiring Color Scheme.

All panels shall have a factor enamel finish.

Conductors for power shall be stranded copper, rated at 75 degrees (C), with insulation suitable for both dry and wet locations. Conductors sizing is to be done in accordance with NEC requirements. Power conductors shall be continuous with no field splices allowed.

Wire size for controls shall be #14 AWG copper stranded rated for 90 degrees (C). Wire size for Supervisory Control and Data Acquisition (SCADA) controls shall be #16 AWG cooper stranded rated for 90 degrees (C).

Due to the potential presence of corrosive gases, greases, oils and other constituents present, all mounting hardware shall be type 316 stainless steel and seal-offs are to be installed in conduits leading to the pump control panels and junction boxes. All enclosures and disconnects shall be padlock lockable.

The electrical disconnect for the NEMA 4X, stainless steel enclosure. Provide with fusing class size based on the characteristics of the motor loads served and the available fault current. Provide a surge arrestor in a separate NEMA 4X, SS316 enclosure.

Electrical equipment shall comply with the latest version of the NFPA National Electrical Code (NEC) requirements for Class 1, Group C and D, Division 1 locations. Additionally, equipment located in wet wells shall be suitable for use in corrosive environments. Flexible cable is to be provided with watertight seals and separate strain relief. High water float switches shall be normally open and will be of the non-mercury type.

All electrical equipment and connections located in wet wells and dry wells shall be rated for Class 1, Division 1 service and shall be explosion-proof.

Any underground electrical conduits shall be gray, rigid nonmetallic conduits. Field-manufactured bends are not acceptable. Only fabricated conduit bends are allowed. Buried conduit shall have a cover depth not to exceed 18 to 24 inches. All exposed conduits shall be rigid aluminum.

Motors shall be controlled with variable frequency drives (VFDs) with communication module as specified by Honeywell.

VFDs must be rated for operation at temperatures equal to 50 degrees (Centigrade) or higher. Consider soft starters in lieu of VFDs; pending review and approval during the pre-design meeting. Coordinate with Mechanical Design.

When VFDs are used, the pump control panel must be insulated and provided with a closed loop, climate controlled with appropriate air conditioning.

Grounding systems shall have a maximum ground resistance per NEC. The Design Engineer will incorporate special soils, such as graphite compounds if needed to improve ground resistance capabilities.

Special considerations for submersible stations include the following:

Design the electrical system, control, and alarm circuits to allow for disconnection outside the wet well. Terminals and connectors shall be protected from corrosion by location outside the wet well in a NEMA 4X stainless steel enclosure.

Locate the motor control center outside the wet well, readily accessible and protected by conduit seals to prevent the atmosphere of the wet well from entering the control center.

Pump motor cables shall meet the requirements of the NEC for cords in industry wastewater pumping stations. Power cord terminal fittings shall be corrosion-resistant and constructed in a manner to prevent moisture entry into the cable. They shall also be provided with strain relief devices.

6.2 Electrical Demolition

The abandonment and modifications of lift stations shall consist of removing all pumps, motors, and electrical appurtenances above finished grade. Government shall have the opportunity to evaluate infrastructure to be salvaged.

Coordinate with other trades to ensure the electrical service is sequenced with other trades to keep the lift station in operation.

6.3 Electrical Construction

Contractor will be required to verify and trace all existing panel sources, main circuit breakers, branch circuits, and existing loads information.

Provide a 120-volt, 20-amp duplex, GFI, receptacle in an “in-use” weatherproof box equipped with a clear cover. Light switches shall also be installed in a weatherproof box with an “in use” clear weatherproof cover.

Power Monitoring: Coordinate with Honeywell for what information and alarms need displayed and logged. As a minimum SCADA shall display the following:

- KWhr, KVARh, KVAh
- 3 phase voltages (per phase and average)
- 3 phase current (per phase and average)
- Frequency (Hz)

Provide general illumination levels equal to 1.0 foot-candle, on average, in lift station equipment & Control Panel area. Use LED for general illumination. Site lighting is not to interfere with surrounding area.

Appendix A

Acronyms and Abbreviations

A	Ampere(s)
AABC	Associated Air Balance Council
ACCA	Air Conditioning Contractors Association
ACI	American Concrete Institute
ACS	Access Control System
ACT	Acoustic Ceiling Tile
AFB	Air Force Base
AFF	Above Finished Floor
AFI	Air Force Instruction
AHJ	Authorities Having Jurisdiction
AHU	Air Handling Unit
AISC	American Institute of Steel Construction
ANSI	American National Standards Institute
ASCE	American Society of Civil Engineers
ASHRAE	American Society of Heating, Refrigeration and Air-Conditioning Engineers
ASTM	American Society for Testing and Materials
BTHU	British thermal unit
BUR	Built-Up Roof
CAC	Common Access Card
CBT	Computer Based Training
CCTV	Closed Circuit Television
CIPS	Cyberspace Infrastructure Planning System
CMU	Concrete Masonry Unit
COMM	Communications
DB	Dry Bulb
DoD	Department of Defense
DX	Direct Expansion
E	East
EBI	Enterprise Building Integrator
EMI	Electromagnetic Interference
EMT	Electrical Metallic Tubing
FCI	Fire Control Instruments
FCU	Fan Coil Unit

FF-L	Federal Specification
FPS	Feet Per Second
FT	Feet
GSA	General Services Administration
HR	Hour(s)
HSS	Hollow Structural Section
HVAC	Heating, Ventilation and Air Conditioning
IAW	In Accordance With
IBC	International Building Code
IC	Intelligence Community
ICD	Intelligence Community Directive
ICS	Intelligence Community Standard
IDS	Intrusion Detection System
IEBC	International Existing Building Code
IECC	International Energy Conservation Code
IMC	International Mechanical Code
IN	Inches
IPC	International Plumbing Code
IT	Information Technology
kVA	Kilovolt-Ampere
LAN	Local Area Network
LED	Light Emitting Diode
MOD	Motorized Damper
N	North
NEBB	National Environmental Balancing Bureau
NFPA	National Fire Protection Association
NIPR	Non-Secure Internet Protocol Router
PSF	Pounds per Square Foot
RF	Radio Frequency
RFI	Radio-Frequency Interference
SIPR	Secure Internet Protocol Router
SMACNA	Sheet Metal and Air Conditioning Contractors' National Association
S	South

STC	Sound Transmission Class
TAFB	Tinker Air Force Base
UFC	Unified Facilities Criteria
UPS	Uninterruptible Power Supply
V	Volts
VAV	Variable Air Volume
W	West
WB	Wet Bulb
WG	Water Gauge
§	Subsection
°F	Fahrenheit

Appendix B

Civil Calculations and Cutsheets

Appendix C

Architectural Calculations and Cutsheets

Appendix D

Structural Calculations and Cutsheets

Appendix E

Mechanical / Plumbing Calculations
and Cutsheets

F-4300 SUBMITTAL AND DATA SHEET

TO:

DATE:

REVISION:

PROJECT NAME:

CONTRACTOR:

ENGINEER:

ONICON REP:

SUBMITTAL FOR:

- ☐ RECORD
- ☐ APPROVAL

APPROVED BY:

RELEASED FOR:

- ☐ MANUFACTURING AND SHIPMENT
- ☐ HOLD FOR RELEASE
- ☐ APPROVED
- ☐ APPROVED AS NOTED
- ☐ DISAPPROVED

EXPLANATION:

PLEASE RETURN APPROVED DRAWINGS TO:

ATTENTION:

SUBMITTED BY:

DESCRIPTION

ONICON F-4300 Clamp-on Ultrasonic Flow Meters/ Thermal Energy Measurement Systems offer an ideal solution for liquid flow and energy measurement in existing systems when it is impractical to install traditional inline or insertion style flow meters. The innovative design incorporates matched precision clamp-on transducers and signal processing circuitry to accurately measure the flow of most liquids over a wide velocity range.

Each F-4300 is provided with transducers and easy-to-use mounting hardware, factory supplied transducer cabling, and a wall mount enclosure with LCD and user interface keypad. When ordered with the BTU (energy) option, the F-4300 is provided with a matched pair of temperature sensors.

Output signals include analog output(s), digital outputs, digital inputs (energy version) and an isolated RS485 output for connecting to BACnet MS/TP or MODBUS RTU networks.

APPLICATIONS

- Chilled water, hot water, condenser water & water/glycol systems for HVAC
- Steam condensate (pumped)
- Domestic/municipal water
- Process water & other clean liquids

CALIBRATION

All F-4300 flow/ energy meters and temperature sensors are wet-calibrated in a flow laboratory against standards that are directly traceable to National Institute of Standards and Technology (N.I.S.T.). A certificate of calibration accompanies every meter.

GENERAL SPECIFICATIONS*

F-4300 TRANSMITTER		
TRANSMITTER PERFORMANCE	ACCURACY	± 1.0% of reading from 1.5 to 40 ft/s (0.46 to 12.2 m/s) (26:1 range) ± 0.015 ft/s (±0.0046 m/s) for velocities below 1.5 ft/s (0.46 m/s)
	REPEATABILITY & LINEARITY	± 0.25%
DIFFERENTIAL TEMPERATURE SENSORS	Temperature sensors meet EN1434/CSA C900.1 accuracy requirements for 1K sensors for cooling applications, 32°F to 77°F Temperature sensors meet EN1434/CSA C900.1 accuracy requirements for 2K sensors for heating applications, 140°F to 212°F	
	ONICON CURRENT BASED TEMP SENSOR**	Precision solid state current based sensors. Signal (mA) is unaffected by wire length. Overall differential temperature measurement uncertainty of ± 0.15°F over the application range
	PT1000 RTD TEMP SENSOR**	1000Ω platinum RTDs calibrated to a differential measurement uncertainty of ± 0.18°F over the stated range
OPERATING CONDITIONS	AMBIENT OPERATING	-5°F to 140°F
INPUT POWER	AVAILABLE OPTIONS	• 24 V AC/DC, 50/60 Hz, 10 VA max • 110-240 VAC, 50/60 Hz, 10 VA max
I/O SIGNAL	ANALOG OUTPUT(S)	Flow Meter: One (1) isolated AO, 4-20 mA or 0-5 VDC BTU Meter Option: Three (3) isolated AOs, 4-20 mA or 0-5 VDC
	DIGITAL OUTPUTS	Flow Meter: Two (2) programmable DOs, SPST, form A, N.O. BTU Meter Option: Six (6) programmable DOs DO 1-2 : SPST, form A, N.O. DO 3-6: SPDT, form C Pulse duration: 500 ms Min. pulse interval: 2s Max. load voltage/current: 50V/ 250mA AC/DC
	DIGITAL INPUTS	BTU Meter Option: Three (3) DIs For use with open collector sinking or contact closure outputs Input voltage logic low: < 0.5 VDC Input voltage logic high range: 5 - 24 VDC Min. pulse interval: 2s
ELECTRONICS ENCLOSURE	NEMA 4X (IP67) polycarbonate enclosure with clear shatter proof cover	
	DISPLAY	Large, easy to read, backlit, 128 x 64 dot matrix display
	DATA LOGGER	Built-in 128 MB data logger with type A USB output. Capacity for 26 million data points
PROGRAMMING	Menu driven via five (5) programming keys	
ELECTRICAL CONNECTIONS	Enclosed terminal blocks, cable access through three (3) ½" conduit openings for signal and power, and one (1) ¾" conduit opening for transducer cables.	
NETWORK CONNECTIONS	AVAILABLE OPTIONS	• RS485 serial interface, BACnet MS/TP (default) or MODBUS RTU • Optional MODBUS TCP/IP (24 VDC only)

* SPECIFICATIONS subject to change without notice.
** Installation hardware are provided separately.

GENERAL SPECIFICATIONS* (CONTINUED)

NETWORK CONFIGURATION & ADDRESSING	BAUD RATES	4800, 9600, 19200, 38400, or 76800
	DEVICE ADDRESS RANGE	1 - 247
	DEVICE INSTANCE RANGE	1 – 4,194,302 (BACnet® only)
	PARITY	None, Even, Odd (MODBUS® RTU only)
APPROVALS	CE	2014/30/EU EMC Directive
F-4300 FLOW SENSOR		
PERFORMANCE	SENSING METHOD	Ultrasonic differential transit time velocity measurement via non-wetted transducers
OPERATING CONDITIONS	FLUID PROPERTIES	Clean liquids in full (pressurized) pipes
	FLUID VELOCITY RANGE	0.07 ft/s to 40 ft/s
	FLUID TEMPERATURE RANGE	-40°F to 250°F
	PIPE MATERIALS	Suitable for use in a wide range of metallic and non-metallic piping systems
	PIPE SIZE RANGE	1/2" through 48", based on transducer series selected.
TRANSDUCER DESIGN - 10 SERIES	OPERATING FREQUENCY	2.56 Mhz
	PIPE SIZE RANGE	1/2" through 4"
	TRANSDUCER HOUSING	CF8M 316 Stainless Steel
	CABLE CONNECTIONS	• Triax cable with BNC style connectors and sealing jacket • Triax cable with NEMA 6 (IP67) direct connection for wet locations
	MOUNTING KIT	304 Stainless Steel mounting brackets with conduit connection, 200 Series Stainless Steel pipe clamps, and alignment and spacer tool
TRANSDUCER DESIGN - 20 SERIES	OPERATING FREQUENCY	1.28 Mhz
	PIPE SIZE RANGE	2" through 10"
	TRANSDUCER HOUSING	CF8M 316 Stainless Steel
	CABLE CONNECTIONS	Triax cable with BNC style connectors and sealing jacket
	MOUNTING KIT	304 Stainless Steel mounting brackets with conduit connection, 200 Series Stainless Steel pipe clamps, and alignment and spacer tool
TRANSDUCER DESIGN - 30 SERIES	OPERATING FREQUENCY	640 kHz
	PIPE SIZE RANGE	12" through 48"
	TRANSDUCER HOUSING	CF8M 316 Stainless Steel
	CABLE CONNECTIONS	Triax cable with BNC style connectors and ½" MNPT conduit connection and NEMA 4 (IP66) threaded strain relief.
	MOUNTING KIT	304 Stainless Steel mounting brackets with conduit connection, 200 Series Stainless Steel pipe clamps, and alignment and spacer tool

* SPECIFICATIONS subject to change without notice.

METER ORDERING INFORMATION

Model F-4300- A B C D - EEFF - GG

A = Electronics Enclosure
1 = NEMA 4X Polycarbonate

B = Input Power
1 = 24 V AC/DC
2 = 110 - 240 VAC

C = Feature Set & I/O
1 = Flow only, one (1) AO, two (2) DO and RS485, BACnet or MODBUS
2 = Flow only, one (1) AO, two (2) DO and MODBUS TCP/IP [1]
3 = Flow and Energy, three (3) AO, three (3) DI, six (6) DO and RS485, BACnet or MODBUS
4 = Flow and Energy, three (3) AO, three (3) DI, six (6) DO and MODBUS TCP/IP [1]

D = Transducer Cable Length
1 = 25' transducer cable, BNC connector[2]
2 = 50' transducer cable, BNC connector[2]
3 = 100' transducer cable, BNC connector[2]
4 = 25' transducer cable, submersible connection (NEMA 6 - IP67)[3]
5 = 50' transducer cable, submersible connection (NEMA 6 - IP67)[3]
6 = 100' transducer cable, submersible connection (NEMA 6 - IP67)[3]
7 = 25' transducer cable, BNC connector, threaded strain relief (NEMA 4 - IP66)[4]
8 = 50' transducer cable, BNC connector, threaded strain relief (NEMA 4 - IP66)[4]
9 = 100' transducer cable, BNC connector, threaded strain relief (NEMA 4 - IP66)[4]

EEFF = Transducer Series & Installation Hardware
1212 = Includes pair of 10 Series transducer, 37 deg. w/ ½" to 4" nom. pipe diameter SS mounting bracket
2X21 = Includes pair of 20 Series transducer, 35 to 41 deg. w/ 2" to 6" nom. pipe diameter SS mounting bracket [5]
2X22 = Includes pair of 20 Series transducer, 35 to 41 deg. w/ 8" to 10" nom. pipe diameter SS mounting bracket [5]
3231 = Includes pair of 30 Series transducer, 37 deg. w/ 12" to 16" nom. pipe diameter SS mounting bracket
3232 = Includes pair of 30 Series transducer, 37 deg. w/ 18" to 48" nom. pipe diameter SS mounting bracket

GG = Temperature Sensor Selection
00 = Flow only
O1 = Matched pair of current (mA) based sensors, CHW/CW range [6]
O2 = Matched pair of current (mA) based sensors, HHW range [6]
R2 = Matched pair of 4 wire PT1000 RTDs, 1/2" to 2 1/2" line size, 32°F to 250°F [6]
R3 = Matched pair of 4 wire PT1000 RTDs, 3" and larger line size, 32°F to 250°F [6]
S6 = Matched pair of PT100 current (mA) based sensors, -4°F to 104°F [6]

CLICK HERE

ONICON Order Form

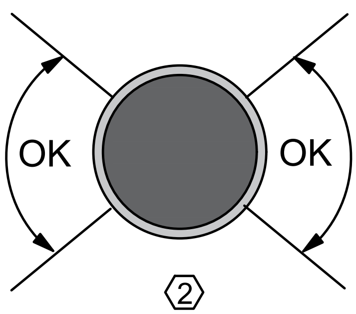
Application and Ordering Guide

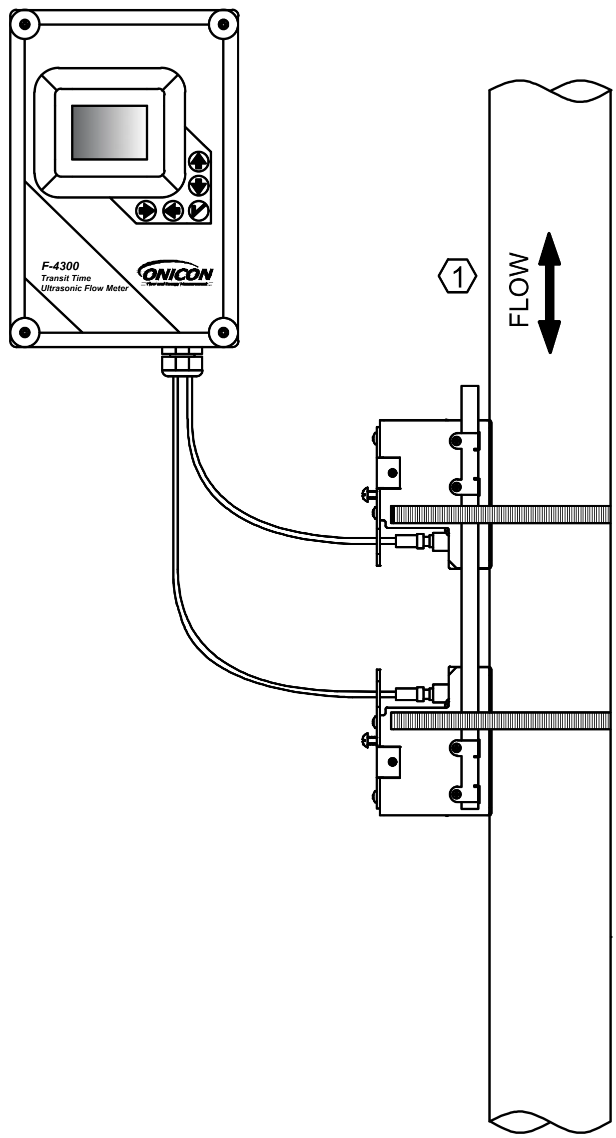
[1] MODBUS TCP/IP requires 24 VDC input power
[2] Only available for transducer series EEFF = 1212, EEFF = 2X21 and EEFF = 2X22
[3] Only available for transducer series EEFF = 1212
[4] Only available for transducer series EEFF = 3231 and EEFF = 3232
[5] Actual transducer selected, 21 through 24, is factory selected at time of order
[6] Only available for feature set C = 3 and C = 4

INSTALLATION CONSIDERATION

A. Flow Meter Orientation

The transducers orientation is one of the most important considerations for accurate flow measurement. Vertical and horizontal pipe positions shown below apply to 10 Series, 20 Series and 30 Series transducers.



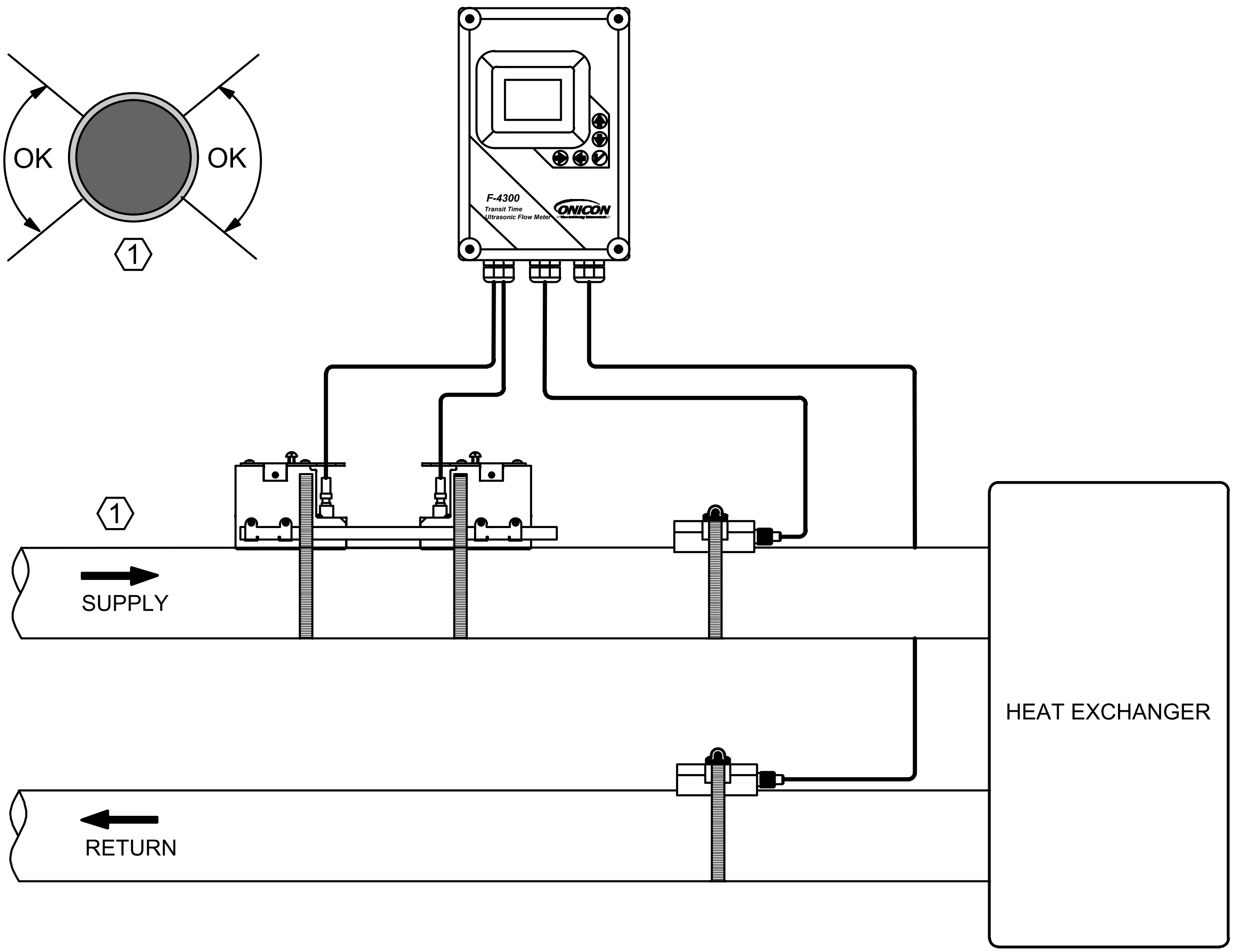


1. Vertical Pipe: Mounting on a vertical pipe is recommended when flow is in the upward direction. When mounting on a vertical pipe flowing in a downward direction, make sure there is sufficient back pressure in the system to maintain a full pipe.
2. Horizontal Pipe: Avoid mounting transducers on the top of a horizontal pipe. The best placement on a horizontal pipe is either the 2:00 to 4:00 or 8:00 to 10:00 positions for two (2) cross (Reflect) or four (4) cross (Double Reflect) mode, or one sensor at 9:00 and one sensor at 3:00 for one (1) cross mode (Direct).

IMPORTANT NOTE
DO NOT mount transducers on the bottom of a horizontal pipe.

INSTALLATION CONSIDERATION (CONTINUED)

B. BTU (Energy) Meter Orientation



1. Horizontal Pipe: Avoid mounting transducers on the top of a horizontal pipe. The best placement on a horizontal pipe is either the 2:00 to 4:00 or 8:00 to 10:00 positions for two (2) cross (Reflect) or four (4) cross (Double Reflect) mode, or one sensor at 9:00 and one sensor at 3:00 for one (1) cross mode (Direct).

   Vertical Pipe: Mounting on a vertical pipe is recommended when flow is in the upward direction. When mounting on a vertical pipe flowing in a downward direction, make sure there is sufficient back pressure in the system to maintain a full pipe.

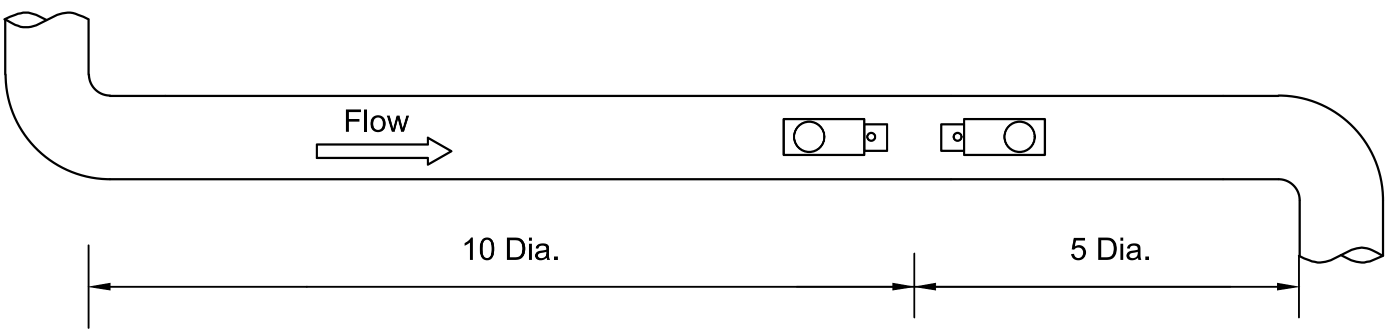
IMPORTANT NOTE
DO NOT mount transducers on the bottom of a horizontal pipe.

INSTALLATION CONSIDERATION (CONTINUED)

C. Straight Run Requirement

For best results, the transducers should be installed on a straight run of pipe free of bends, tees, valves, transitions, insertion probes and obstructions of any kind. For most installations, ten straight unobstructed pipe diameters upstream and five diameters downstream of the transducers is the minimum recommended distance for proper operation. Additional considerations are outlined below.

IMPORTANT NOTE
In some cases, longer straight runs may be necessary where the transducers are placed downstream from devices which cause unusual flow profile disruptions or swirl (for example, modulating valves, two elbows in close proximity and out of plane, etc.).

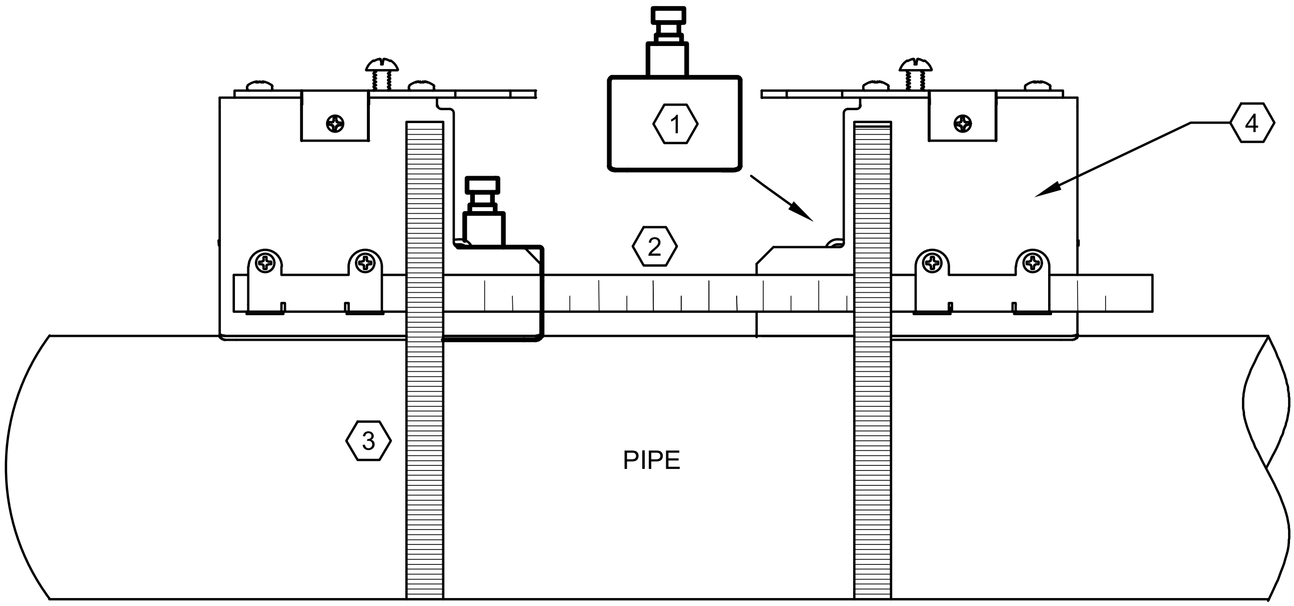


- Avoid installing the transducers downstream from a throttling valve, a mixing tank, the discharge of a positive displacement pump, or any other equipment that could possibly aerate the liquid. The best location will be as free as possible from flow disturbances, vibration, sources of heat, noise, or radiated energy.
- Avoid mounting the transducers on a section of pipe with any external scale. Remove all scale, rust, loose paint, etc., from the location prior to mounting the transducers.
- Do not mount the transducers on a surface aberration (pipe seam, etc.).
- Do not mount transducers from different ultrasonic flow meters on the same pipe.
- Do not run the transducer triaxial cables in common bundles with cables from other instrumentation. You can run these cables through a common conduit ONLY if they originate at the same flow meter.
- Never mount transducers under water.
- Avoid mounting transducers on the top of a horizontal pipe. The best placement on a horizontal pipe is either the 8:00 to 10:00 or 2:00 to 4:00 position for 2 cross (Reflect) or 4 cross (Double Reflect) mode, or one sensor at 9:00 and one sensor at 3:00 for 1 cross mode (Direct).
- Do not mount transducers on the bottom of a horizontal pipe.
- Mounting on a vertical pipe is the recommended installation method if flow is in the upward direction. When mounting on a vertical pipe flowing in a downward direction, make sure there is sufficient back pressure in the system to maintain a full pipe.

TRANSDUCER INSTALLATION SCHEMATICS

A. Overview

Each F-4300 Ultrasonic Flow Meter is provided with a pair of matched ultrasonic transducers. The transducers are mounted (clamped) on to the outside wall of the pipe. Triaxial cables convey the transducer signals to the wall mount enclosure containing the signal processing circuitry and the user interface display.



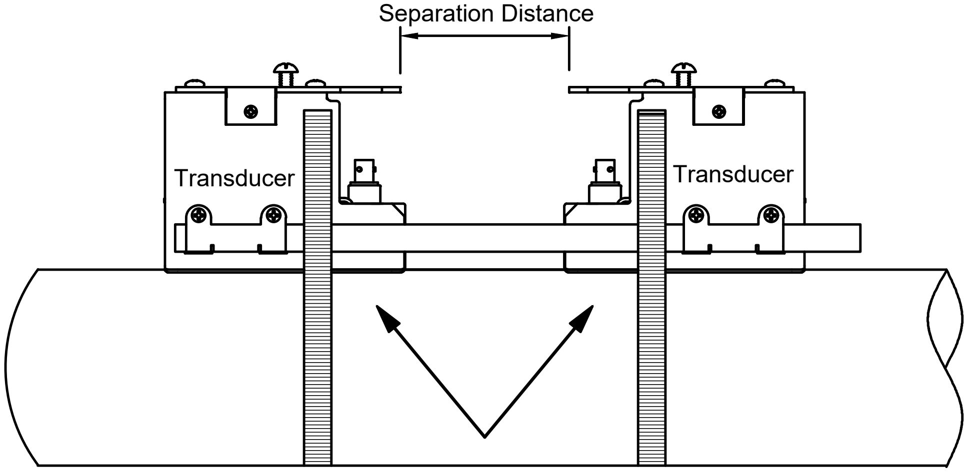
1. Transducer
2. Alignment and spacer tool
3. Adjustable stainless steel pipe clamp
4. Transducer mounting bracket

Ultrasonic transducers can be configured to operate in either 1 (Direct), 2 (Reflect) or 4 (Double Reflect) cross operating modes. The choice of operating mode is dictated by the configuration settings programmed into the meter. For new installations, configuration data is programmed into the meter prior to shipment. Programming data determines the transducer operating mode and the spacing between the transducers.

TRANSDUCER INSTALLATION SCHEMATICS (CONTINUED)

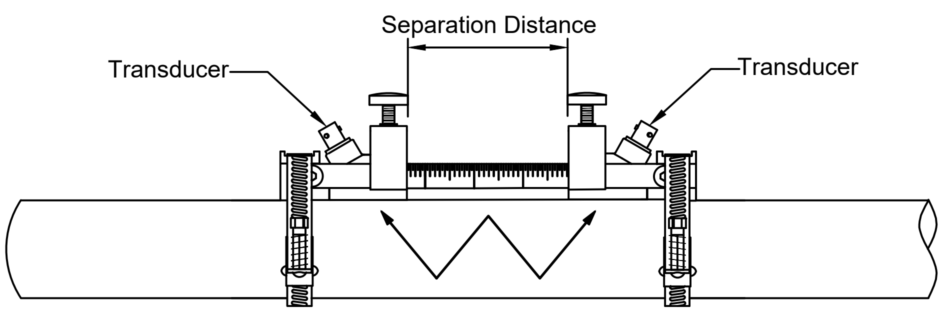
B. Two (2) or Four (4) Cross Mode (Reflect, Double Reflect)

Two (2) cross (Reflect) mode is the recommended operating mode whenever possible. It is the simplest way to mount the transducers. Operating in the reflect mode also minimizes the effects of some flow distortions.



2 Cross Separation Distance (Reflect)

Four (4) cross (Double Reflect) mode is used for some installations in small pipe sizes using the 10 Series transducers.

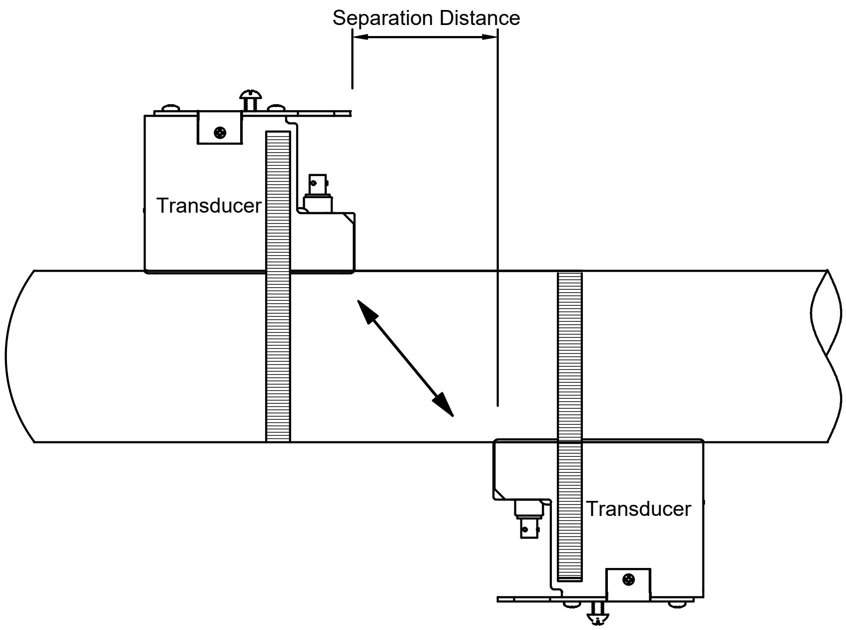


4 Cross Separation Distance (Double Reflect)

TRANSDUCER INSTALLATION SCHEMATICS (CONTINUED)

C. One (1) Cross Mode (Direct)

One (1) cross mount provides a shorter sonic path. The shorter path typically improves performance with difficult pipe conditions, such as older and/ or corroded piping. One (1) cross (Direct) mounting requires half the distance between transducers when compared to the two (2) cross (Reflect) mode and may be the only option if the availability of the mounting space is limited.



1 Cross Separation Distance (Direct)

THERMOWELLS INSTALLATION SCHEMATICS

A. Overview

ONICON offers four different types of temperature sensor pairs and the associated thermowells for use with the F-4300.

CAUTION
Temperature sensor thermowells must match the sensor diameter. Using the wrong diameter hardware will result in significant temperature measurement errors.

Sensor Type	Sensor Diameter	Nominal Pipe Size Range (inches)
ONICON fixed range temperature sensor pair	0.25"	½ to 48"
100 Ω Platinum RTD pair, w/4-20mA transmitters	0.25"	½ to 48"
1000 Ω Platinum RTD pair, 4 wire	5mm	½ to 2½"
1000 Ω Platinum RTD pair, 4 wire	6mm	3 to 48"

The two temperature sensors must be located so they accurately measure only the temperature of the supply line entering and the return line leaving the portion of the piping system for which the energy measurement is being made.

If possible, find an easily accessible location where field wiring connections can be made from floor level. This will facilitate future service. Place the temperature sensors away from strong sources of electrical noise that might affect the performance of the sensors.

One temperature sensor thermowell will need to be placed in the same pipe with the flow meter. If it is an immersion thermowell, it must be located at least five pipe diameters downstream of the flow meter leaving enough clearance to remove either sensor from the pipe without interference from the other sensor.

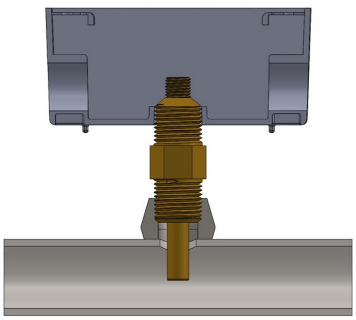
THERMOWELLS INSTALLATION SCHEMATICS (CONTINUED)

B. 0.25” Diameter Thermowell

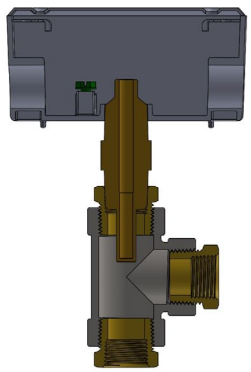
1. Dry Tap Thermowell

Dry tap thermowells are for new construction or scheduled shutdown. The most common installation methods are shown below. Consult ONICON for special applications.

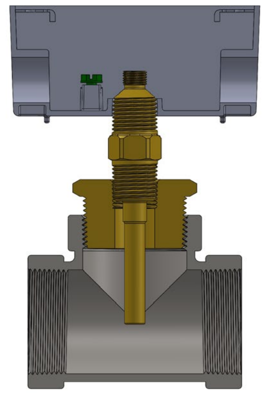
Welded Pipe



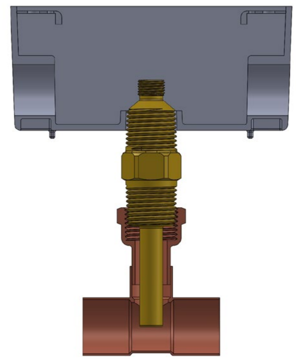
Threaded Tee (for 1/2" pipes)



Threaded Tee (for 3/4" and larger pipes)



Copper Tee



NOTES

1. Thermowell length varies with pipe size.
2. Do not use multiple bushings to reduce the outlet size on threaded tees.

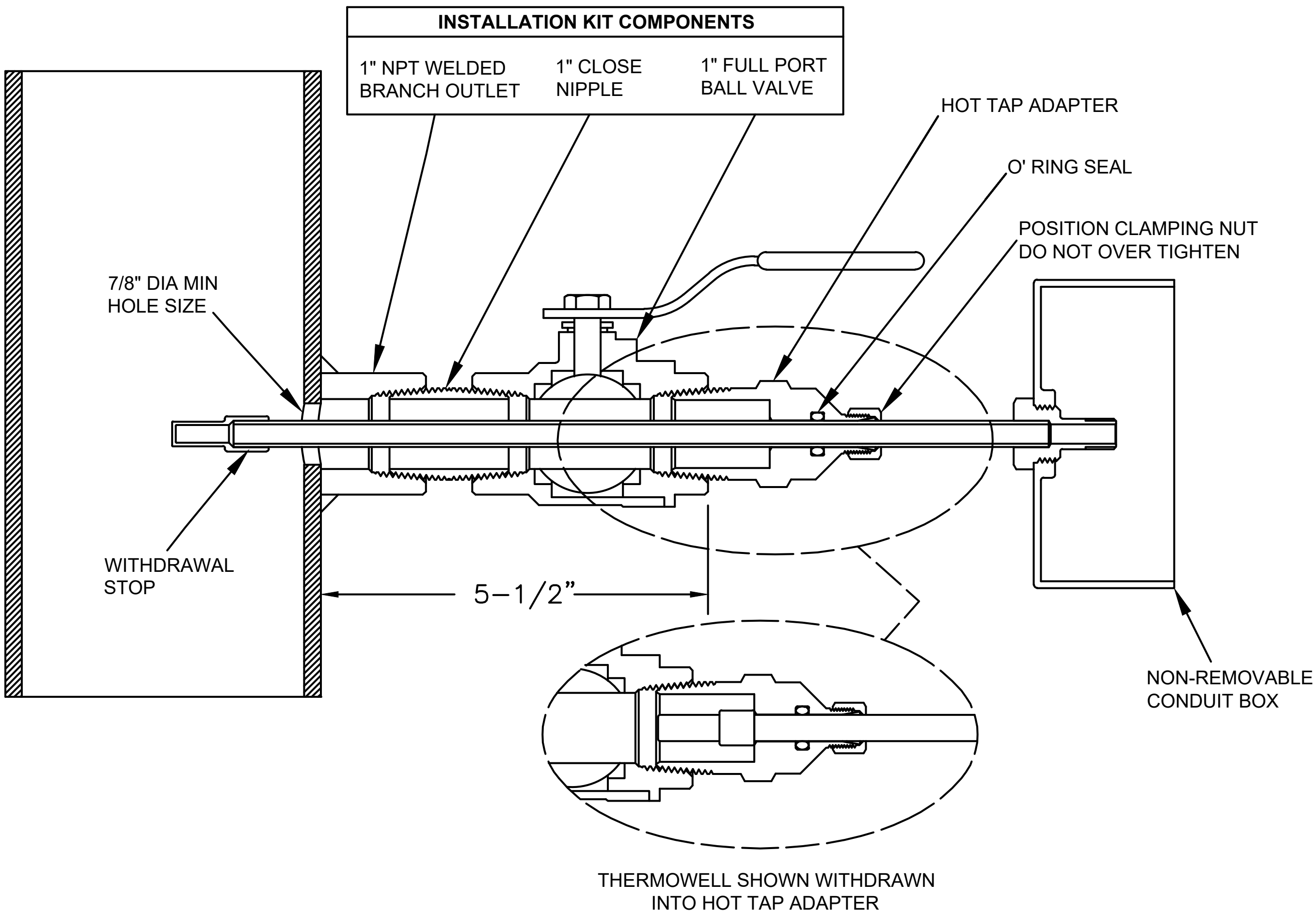
THERMOWELLS INSTALLATION SCHEMATICS (CONTINUED)

B. 0.25" Diameter Thermowell (Continued)

2. Hot Tap Thermowell

Hot tap thermowells are designed for retrofit applications where it is not practical to isolate and drain the pipe section prior to installation. The thermowell is installed through a 1" full port ball valve as shown in the drawing below. A hot tap drilling machine equipped with a 7/8" drill is required to perform this type of installation.

Once the valve assembly has been installed and the hole has been drilled, the thermowell can be inserted into the flow stream without a system shutdown.

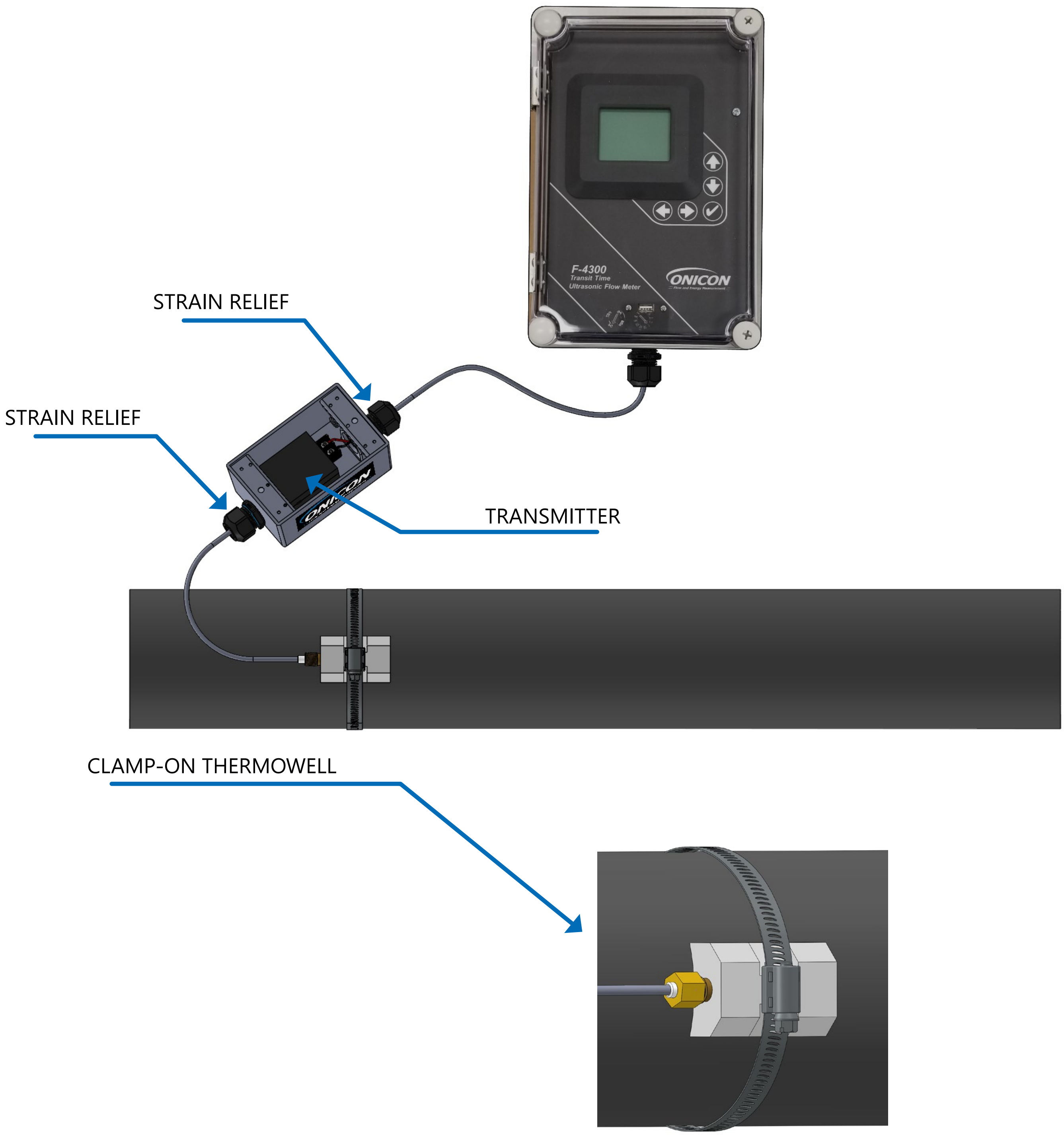


Hot Tap Installation Detail For Thermowell In Welded Pipe

THERMOWELLS INSTALLATION SCHEMATICS (CONTINUED)

C. Clamp-On Thermowell

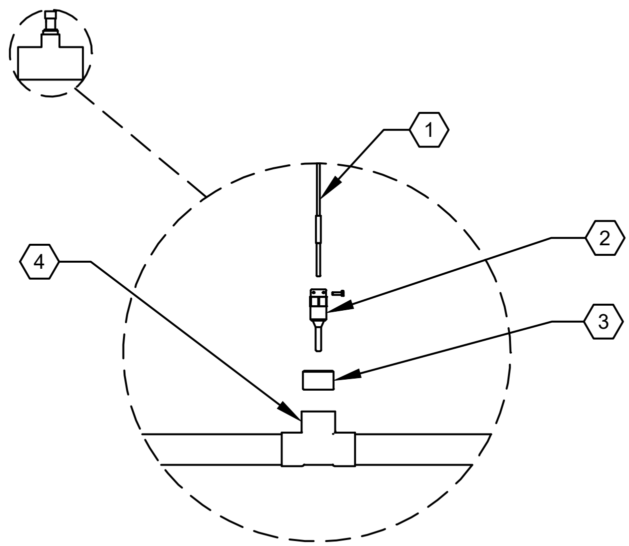
The two clamp-on temperature sensors must be located such that they only measure the temperature of the supply pipe entering and the return pipe leaving the portion of the piping system for which the energy measurement is being made. Once installed, the pipes and clamp-on thermowells must be fully insulated.



THERMOWELLS INSTALLATION SCHEMATICS (CONTINUED)

D. 5mm Diameter Thermowell

5 mm RTDs are provided with thermowells with ½" male NPT process connections. They are designed for use in ½" to 2½" line size tees provided by the customer. The RTDs are push-in type and are held in place with a set screw. Depending on the pipe material, the kit may include a copper sweat bushing or a threaded reducer bushing.

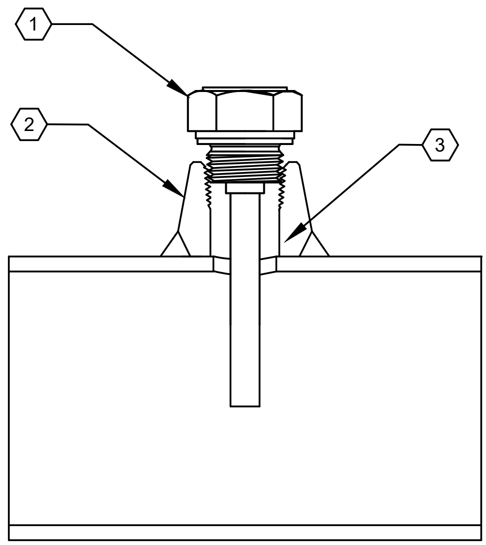


1. RTD temperature sensor – provided by ONICON.
2. 5 mm diameter thermowell – provided by ONICON.
3. 1" solder x ½" NPT bushing OR line size x ½" bushing – provided by customer or ordered from ONICON.
4. Customer supplied line size tee.

THERMOWELLS INSTALLATION SCHEMATICS (CONTINUED)

E. 6mm Diameter Thermowell

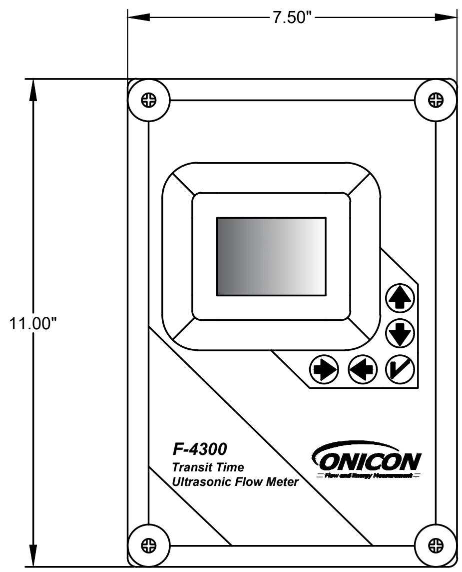
6mm RTDs are provided with matching length thermowells with ½” male NPT process connections. They are designed for use in 3” and larger diameter pipes. The RTDs are push-in type and are held in place with a set screw. The kit includes two (2) weld-on branch outlets with ½” NPT threads.



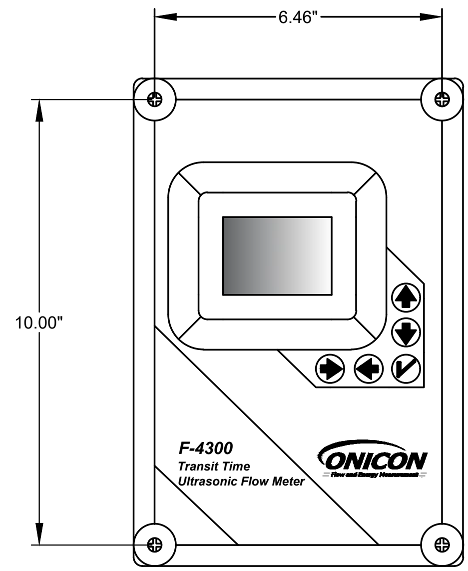
1. 6 mm diameter thermowell - provided by ONICON.
2. ½" NPT weld on branch outlet - purchased from ONICON.
3. ¾" minimum hole size.

DIMENSIONS

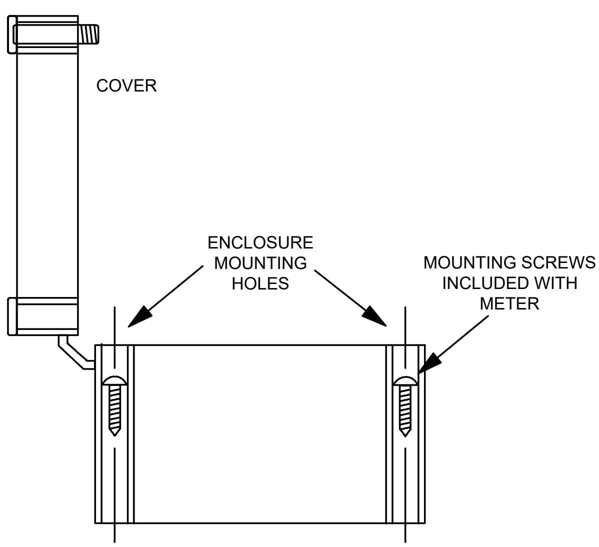
A. Enclosure Dimensions



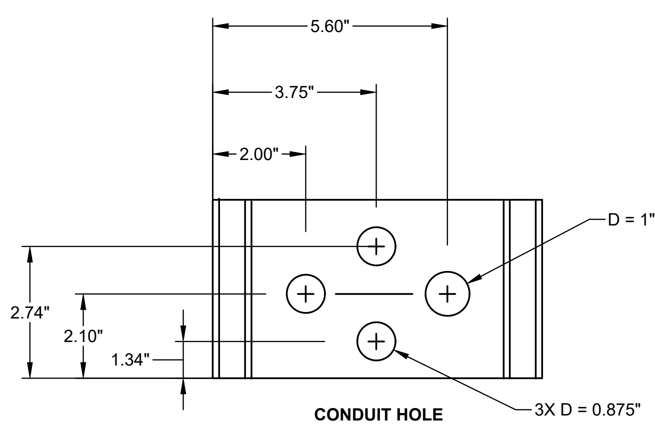
ENCLOSURE DIMENSIONS



MOUNTING HOLD DIMENSIONS



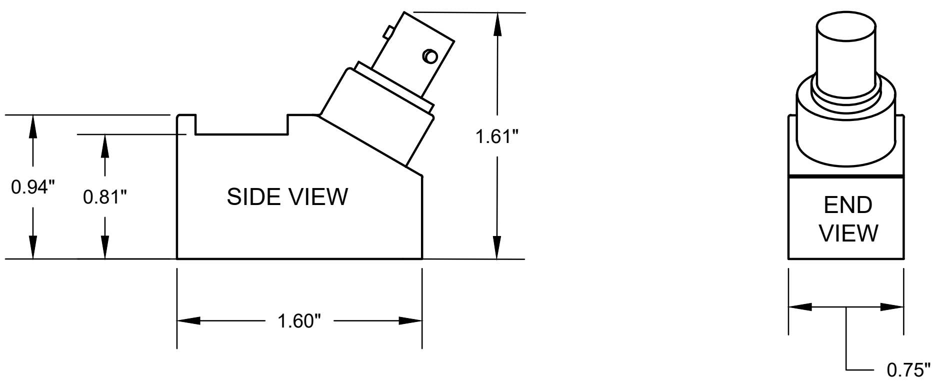
END VIEW

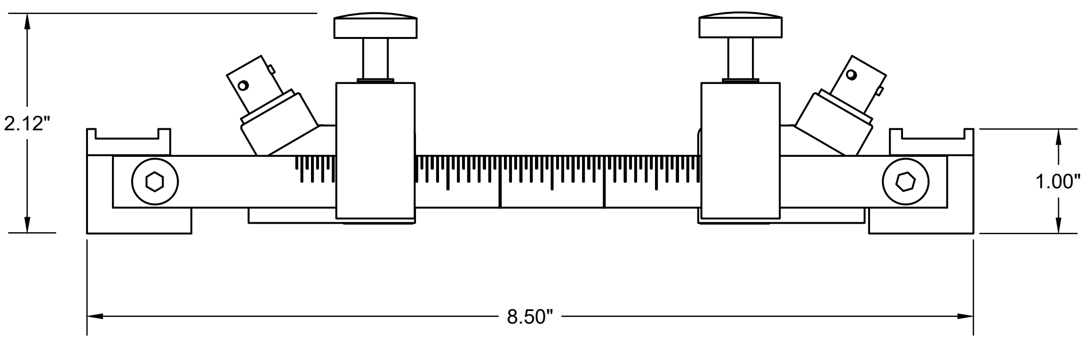


CONDUIT HOLE AND DIMENSIONS

DIMENSIONS (CONTINUED)

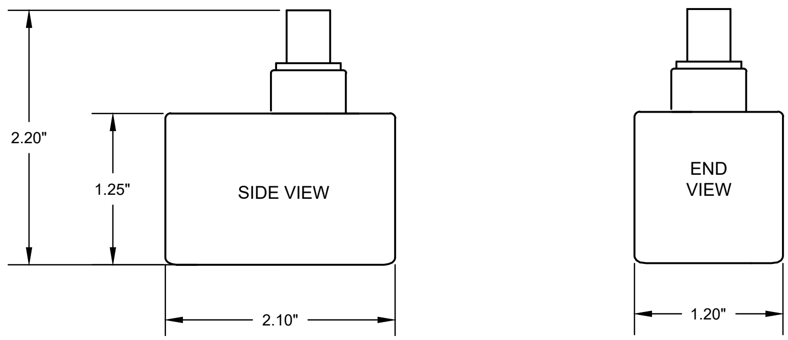
B. 10 Series Transducer Dimensions

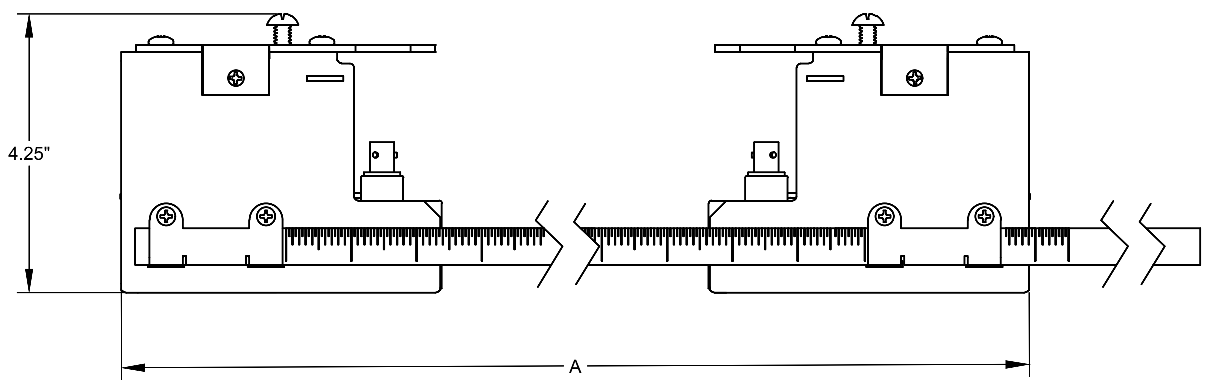




DIMENSIONS (CONTINUED)

C. 20 Series Transducer Dimensions



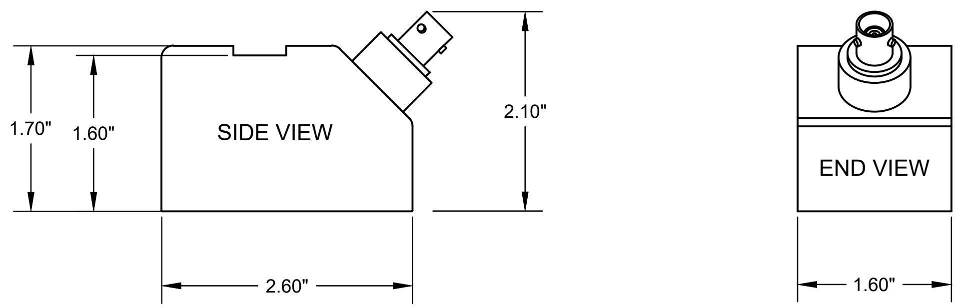


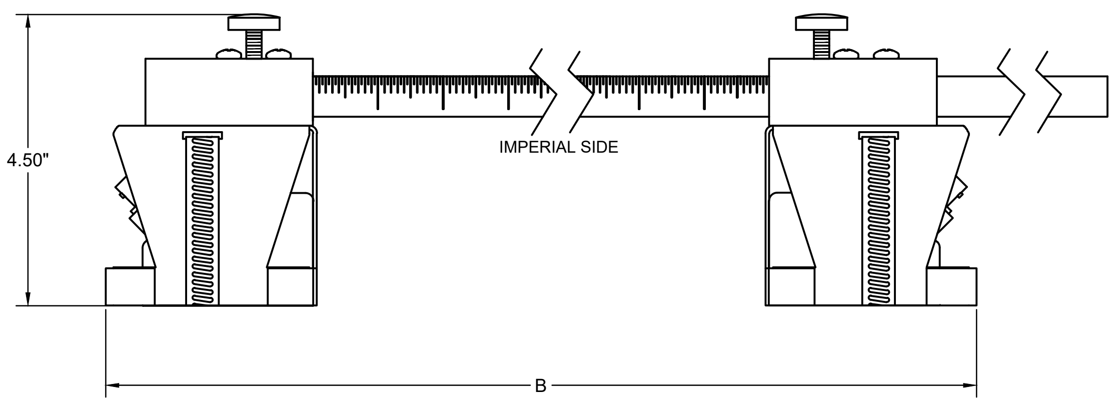
Dimension (A) varies based on pipe material and wall thickness

Pipe Size Range	Dim (A)
2″ to 6″	14″
8″ to 10″	19″

DIMENSIONS (CONTINUED)

D. 30 Series Transducer Dimensions

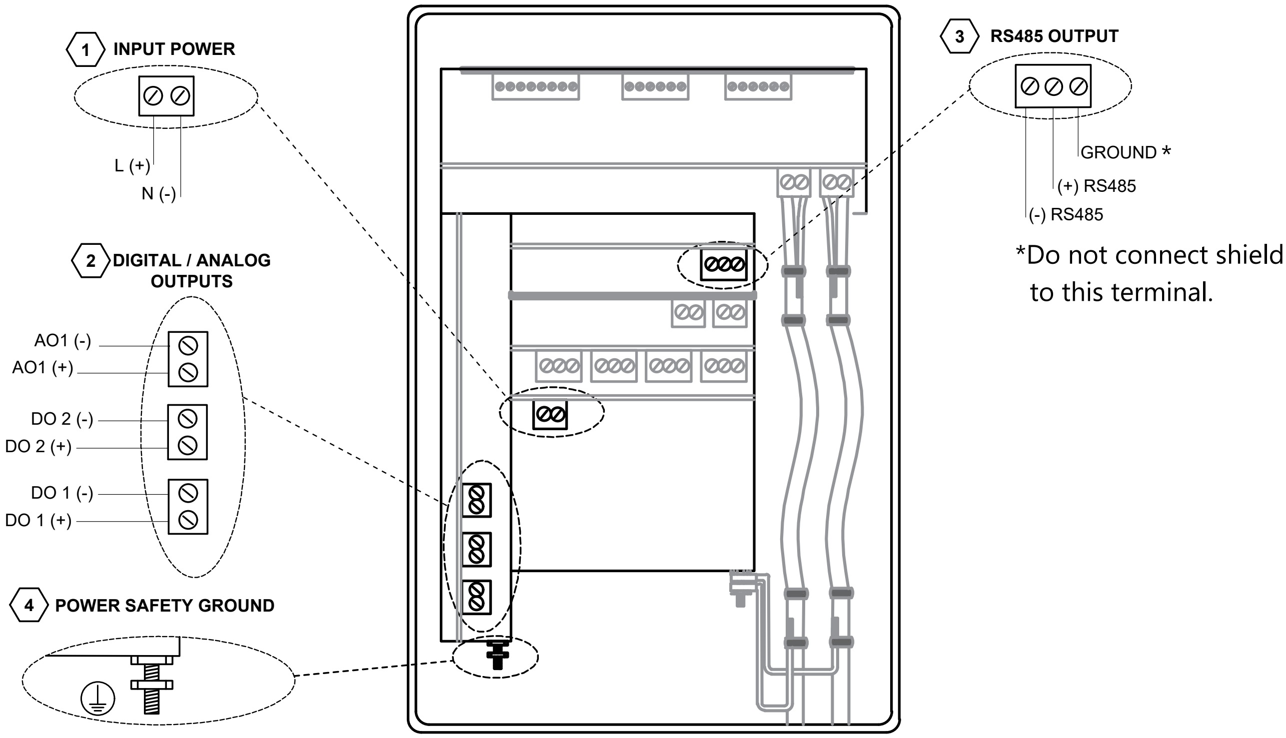




Dimension (B) varies based on pipe material and wall thickness

Pipe Size Range	Dim (B)
12″ to 16″	18″
>16″ to 24″	24″
>24″ to 48″	43″

POWER AND OUTPUT SIGNAL WIRING CONNECTIONS

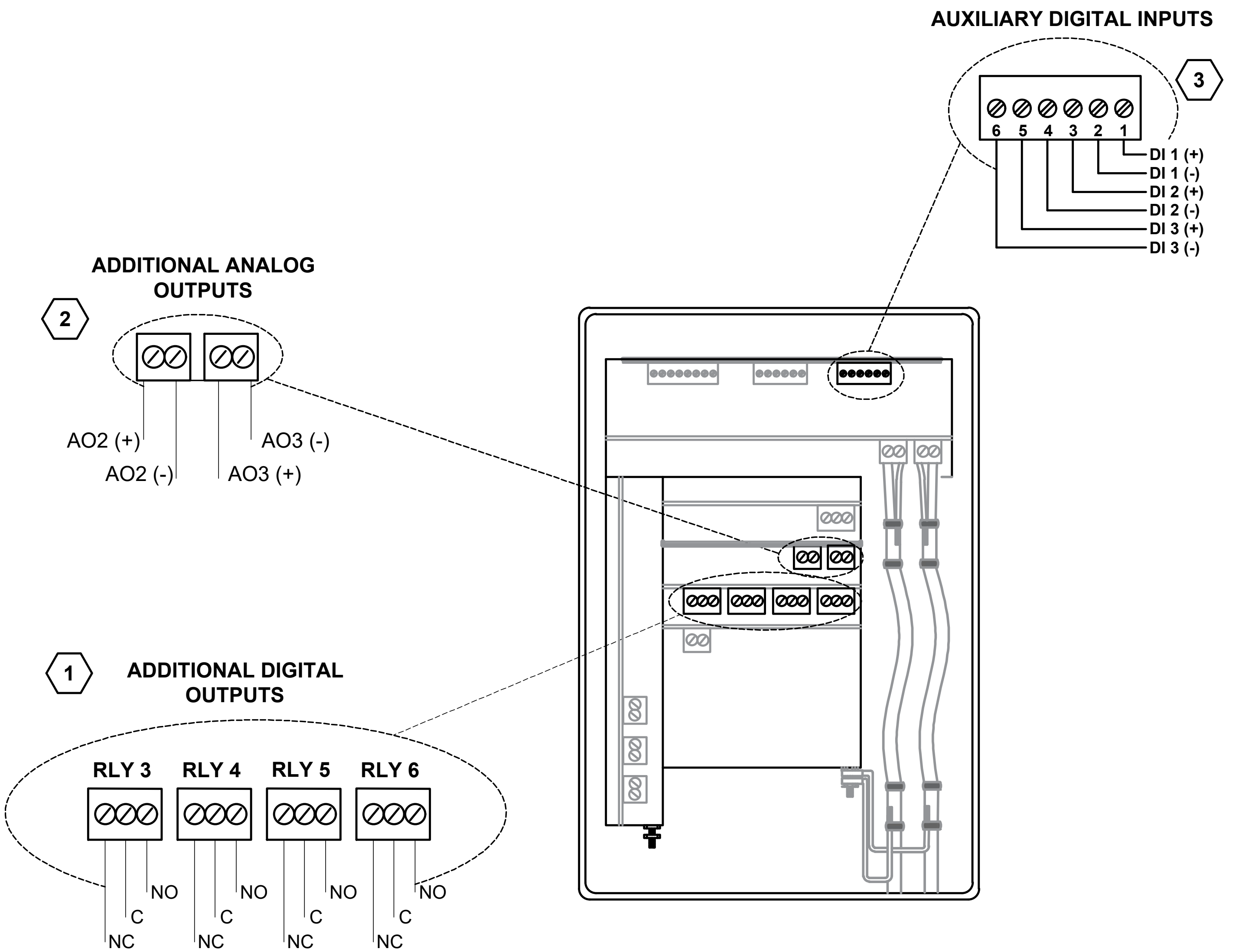


1. Input power
   - 24 VAC/ VDC, 10 VA max OR
   - 110-240 VAC, 50/ 60 Hz, 10 VA max
2. Output configuration
   - Two (2) programmable digital outputs and one (1) active analog output, 4-20 mA or 0-5 VDC (Menu selectable).
3. RS485 BACnet MS/TP or MODBUS RTU.
4. Power safety ground connection.

Power and Output Signal Wiring Notes

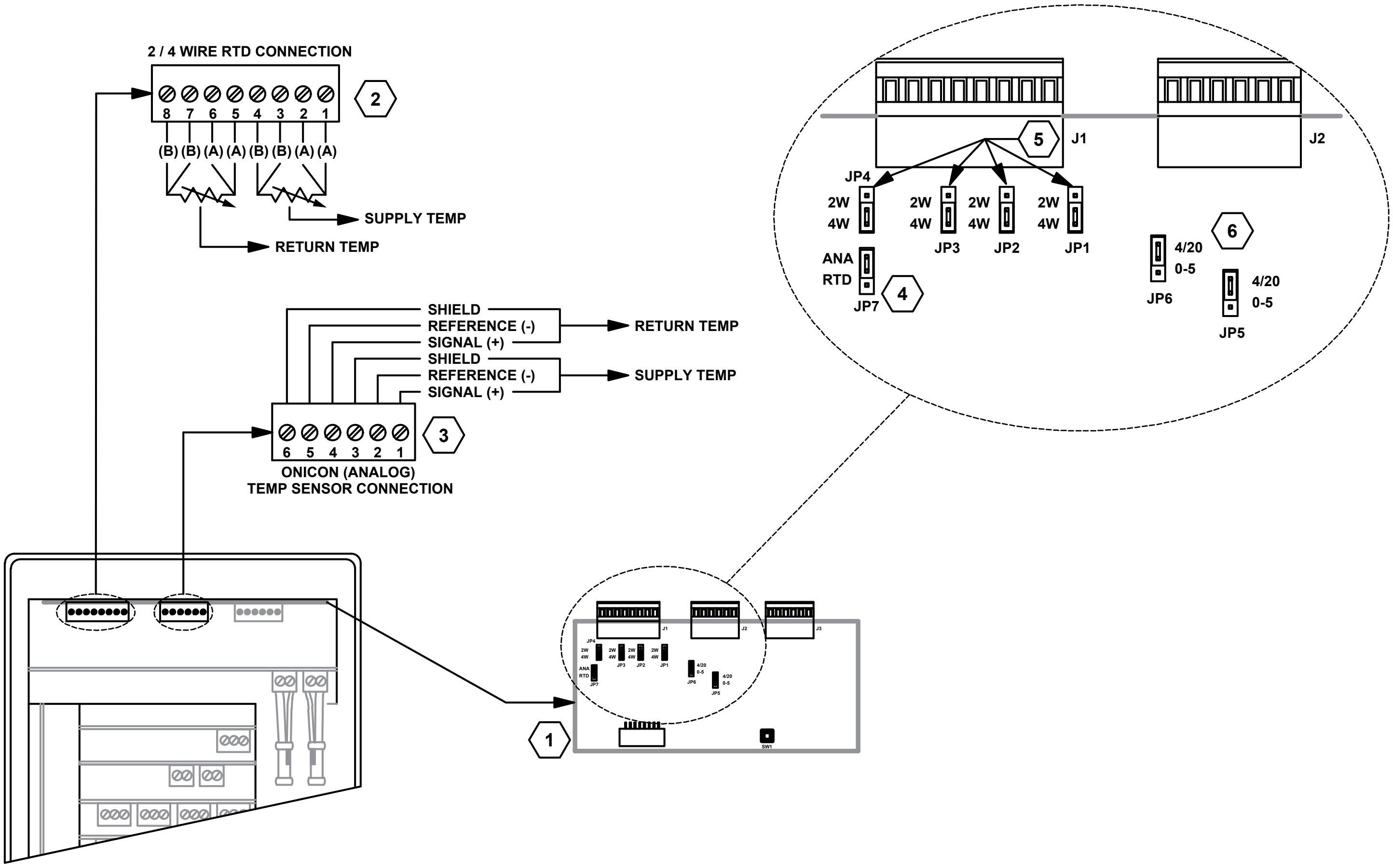
1. If the F-4300 meter will be powered by 24VDC ensure that the input power polarity is correct to avoid polarity issues on the RS485 outputs.
2. Adhere to official BACnet guidelines for selecting data wiring and grounding and implementation, ASHRAE Standard 135 - "BACnet - A Data Communication Protocol for Building Automation and Control Networks".
3. A single grounding path should connect the RS485 ground port on each F-4300 and any third-party devices, ending at a reliable grounding location, such as a grounding rod or a panel's grounding port.
4. The shield of each segment of cable should form a single path, ending either at a grounding rod, electrical panel, or at the controller's grounding point if it is properly grounded. The farthest end of the shield should remain ungrounded (open). DO NOT connect the shield to the RS485 ground port of any F-4300.
5. RS485 Ground should only be shared across the RS485 Ground path with other devices, terminated at the controller, and may not be grounded to the F-4300 chassis or power safety ground. Incorrect grounding may create communications disruptions from grounding loops.
6. Ensure the power safety ground for the F-4300 is effectively connected to a grounding panel or rod.
7. When integrating any ONICON devices into an existing network, check and correct any reversed polarity found in the data channel.
8. Review the BACnet Protocol Implementation Conformance Statement in the manual on page 56 to make sure the BMS and controller stay within property limits when interfacing with the F-4300.

BTU (ENERGY) INPUTS & OUTPUTS WIRING CONNECTIONS



1. Configurable digital outputs terminal block connections
2. Configurable analog outputs terminal block connections (4-20 mA)
3. Configurable digital inputs terminal block connections

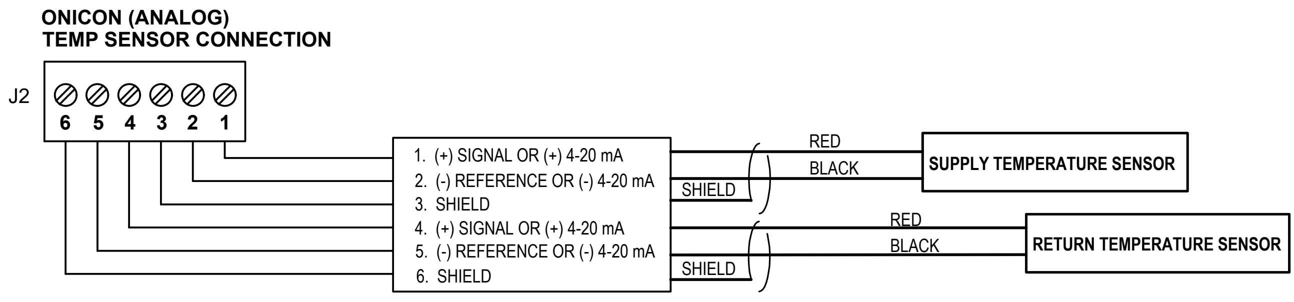
TEMPERATURE SENSOR WIRING CONNECTIONS



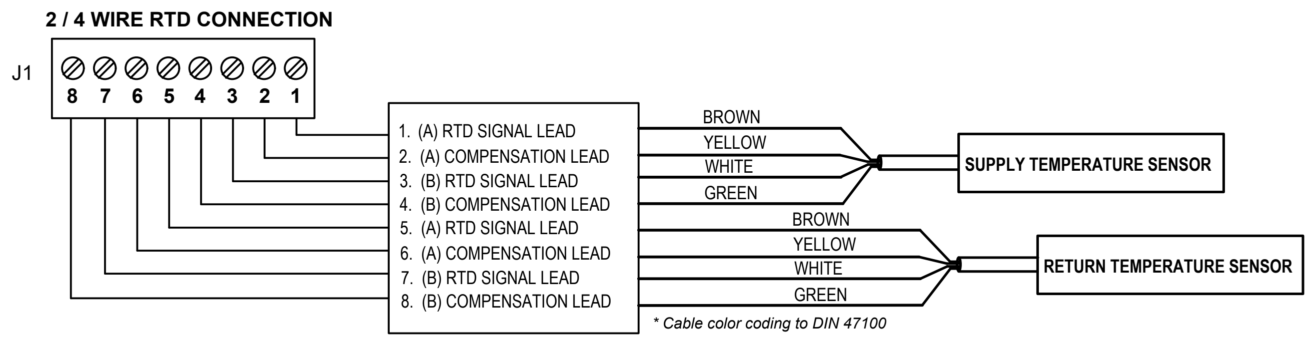
1. Energy computer circuit board.
2. 2 wire / 4 wire RTD connection terminal block.
3. ONICON temperature sensor connection terminal block.
4. Temperature sensor type selection jumper, JP7.
5. RTD type jumper selection, JP1 through JP4. Selects 2 wire vs 4 wire RTD input.
6. ONICON temperature sensor type selection jumpers, JP5 and JP6, 4-20 mA passive input or 0-5 VDC.

TEMPERATURE INPUTS WIRING DETAILS

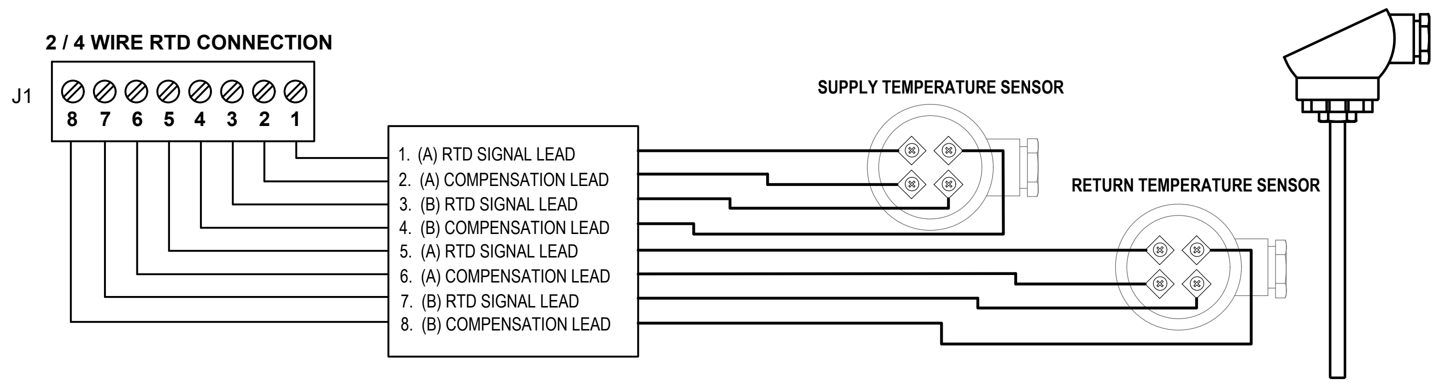
ONICON FIXED RANGE or LOOP POWERED 4-20mA SCALABLE RANGE TEMPERATURE INPUTS



1000 OHMS 4-WIRE PLATINUM RTD TEMPERATURE INPUTS w/ ATTACHED PIGTAIL CABLE



1000 OHMS 4-WIRE PLATINUM RTD TEMPERATURE INPUTS w/ INTEGRAL CONDUIT READY JUNCTION BOX



Refer to Temperature Sensor Wiring Connections on page 24 for additional details on the terminal location.

xylem
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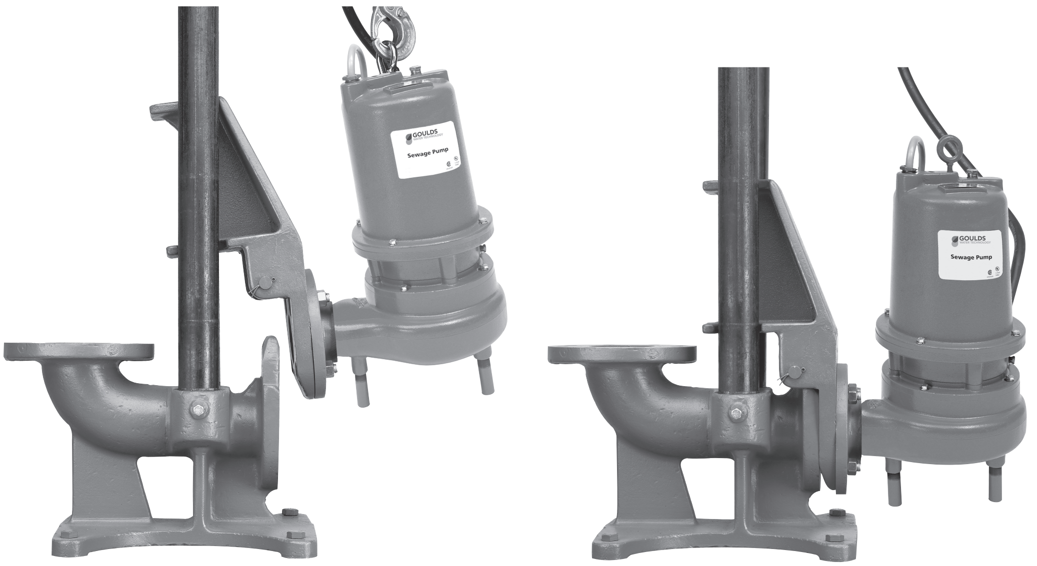
TECHNICAL BROCHURE
BCPCGR R16



Guide Rail Systems

EFFLUENT AND SEWAGE

GOULDS
WATER TECHNOLOGY
a xylem brand

FEATURES

A10-30, 3" X 4" RAIL SYSTEM: Connects to any pump with a 3", 150# ANSI flanged discharge. Outlet is a 4" flanged discharge.

A10-40, 4" X 4" RAIL SYSTEM: Connects to any pump with a 4", 150# ANSI flanged discharge. Outlet is a 4" flanged discharge.

A10-60, 4" X 6" RAIL SYSTEM: Connects to any pump with a 4", 150# ANSI flanged discharge. Outlet is a 6" flanged discharge.

ALL MODELS:

Cast iron construction for standard applications.

Optional brass pump adapter for applications requiring a non-sparking disconnect.

Standard kit contains a base, a pump adapter with all required bolts and fittings, and the upper guide rail positioning bracket.

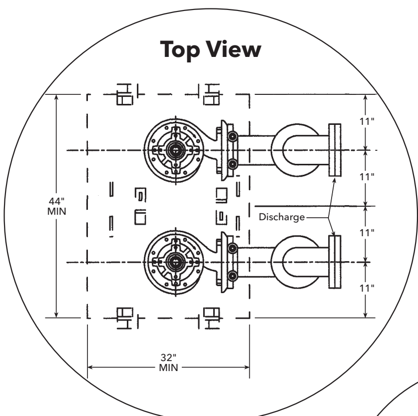
Optional intermediate guide rail brackets are available in either steel or brass for non-sparking applications.

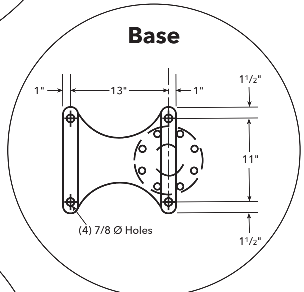
Guide rails are not supplied - they may be sourced locally - 2" stainless steel guide rails recommended.

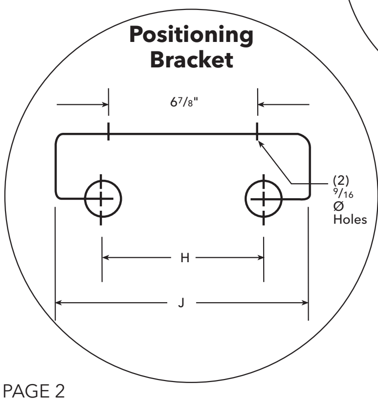
Spare pump adapter kits are available for those who want a back-up pump/adapter ready for an emergency quick change.

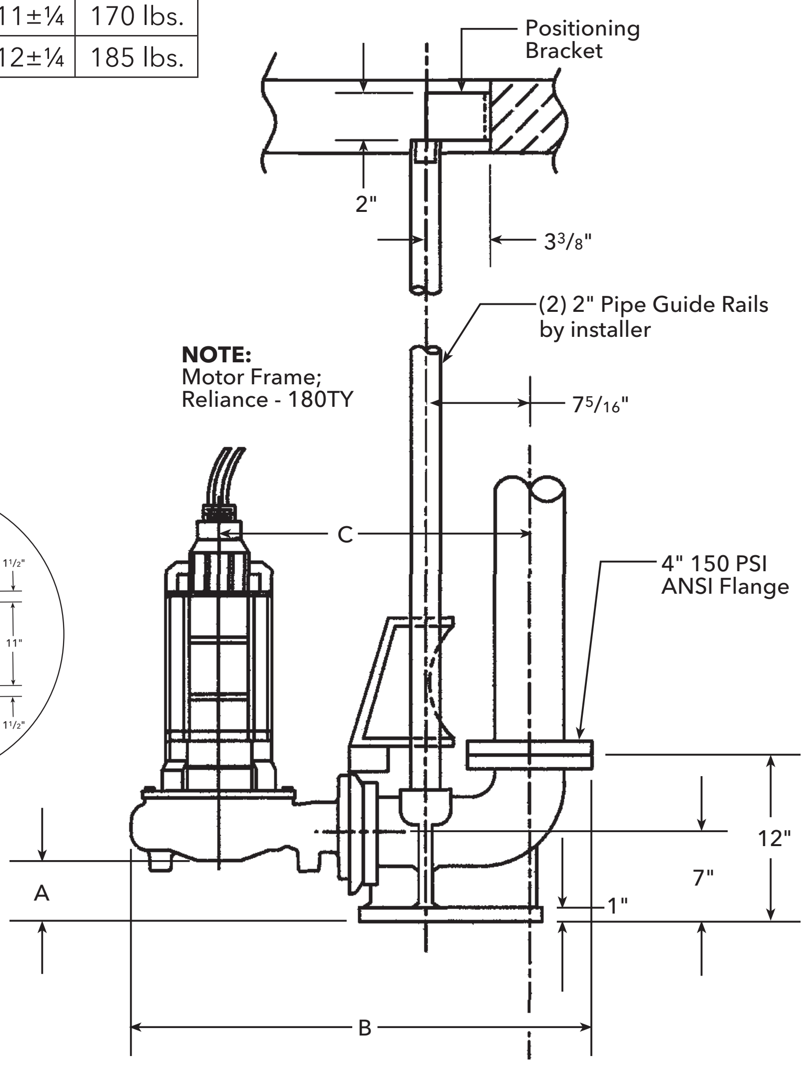
3" AND 4" DISCHARGE GUIDE RAIL SYSTEM

Pump Discharge	Part Number	A	B Max.	C	H	J	Weight
3"	A10-30	4 9/16	33 3/8	22½	6¾	11±¼	170 lbs.
4"	A10-40	3 13/16	34¼	23	7¾	12±¼	185 lbs.









3" AND 4" DISCHARGE GUIDE RAIL SYSTEM

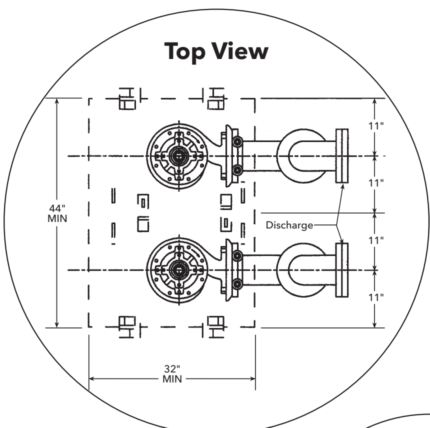
- Heavy duty cast iron construction.
- Twin guide rails provide positive alignment with base.
- No sealing devices required – pump weight provides sufficient force for proper seal.
- Self cleaning design. When pump flange engages base, the shearing action wipes the sealing surfaces clean.

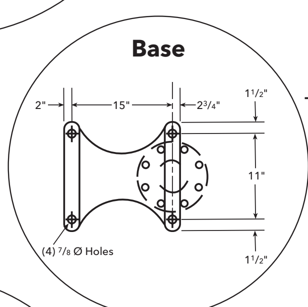
- **System Components Include:**
  - Base with integral cast elbow.
  - Pump adapter – guide assembly with fasteners.
  - Upper guide rail positioning bracket. Carbon steel bracket available as an option in stainless steel.

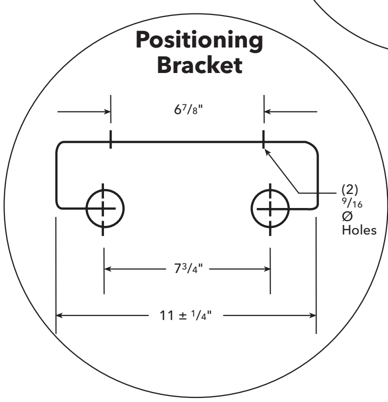
NOTE: Guide rails are not furnished by Goulds Water Technology. Lifting chains and bails need to be ordered separately. Intermediate bracket available as seen on page 4 for pits over 11 feet.

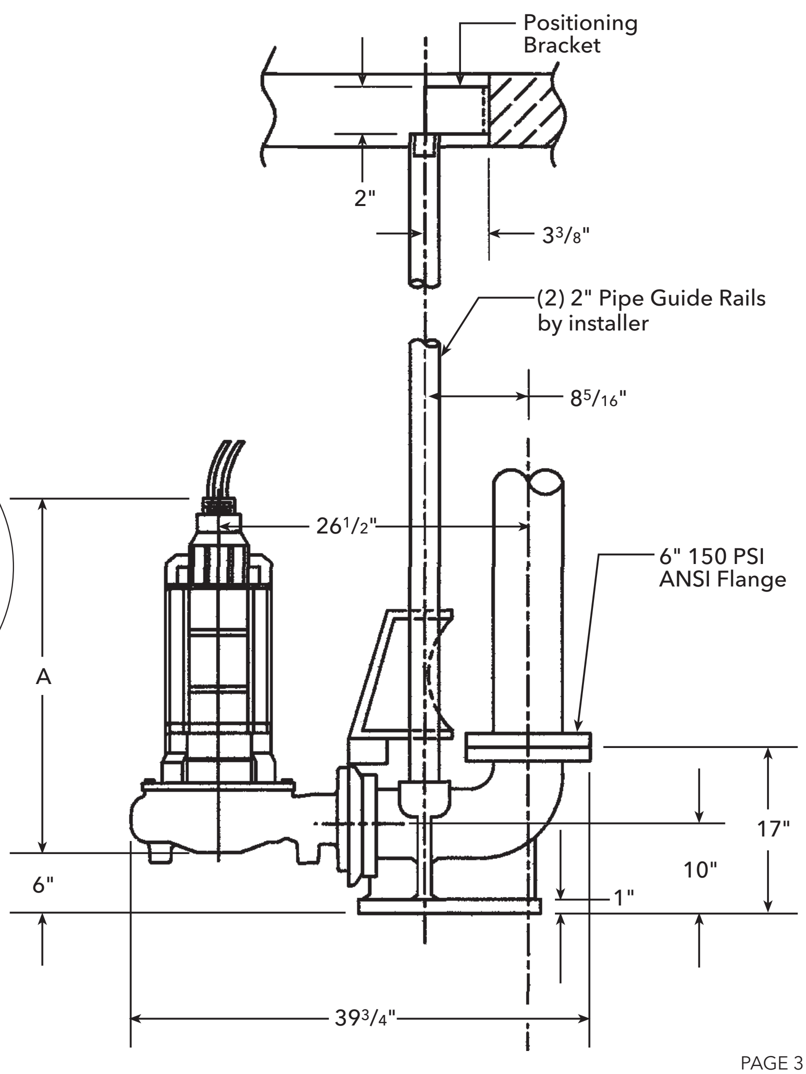
4" DISCHARGE GUIDE RAIL SYSTEM

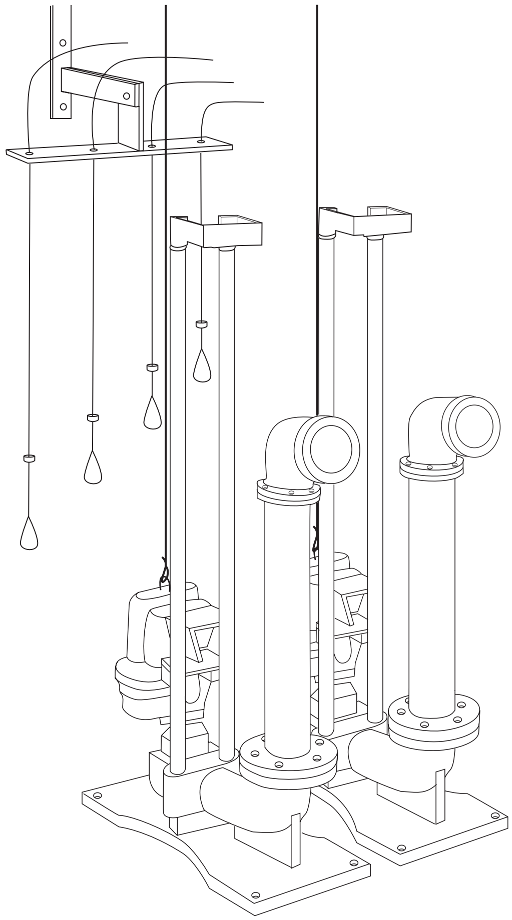
Frame	Pump Discharge	Part Number	A	Weight
210	4"	A10-60	37¾	185 lbs.
250	4"	A10-60	43¹/₈	185 lbs.











PUMP ADAPTER KITS

1K340 – for A10-30 iron
1K341 – for A10-40 / A10-60 iron
1K447 – for A10-30B brass
1K448 – for A10-40B / A10-60B brass

Part numbers are for repairs, component is included in the A10-30, 40 accessory.

INTERMEDIATE GUIDE RAIL BRACKET

A10-30 (B) standard	4K436
A10-40 (B), 60 (B) standard	4K437
A10-30 304 SS	4K631
A10-40 304 SS	4K632

Used on pits over 11 feet for extra support.
Must be purchased separately.

MINIMUM BASIN DIAMETER

	Minimum	Recommended
Simplex	36"	42"
Duplex	48"	60"

UPPER GUIDE RAIL BRACKET

A10-30 (B)	4K467
A10-40 (B), 60 (B)	4K468

Pump Discharge Size	Order Number	ANSI Flanged Discharge Size	Material of Positioning Bracket	Used On These Pumps
3"	A10-30	4" 150 lb. ANSI	Carbon Steel	3WDA, 3DWS, 3WS, 3888D3, 3SD
3"	A10-30SS		Stainless Steel	
4"	A10-40		Carbon Steel	4WDA, 4DWS, 4DWC, 4WS, 3888D4, 4SD, 4NS
4"	A10-40SS		Stainless Steel	
4"	A10-60SS	6" 150 lb. ANSI		
3" XP	A10-30B	4" 150 lb. ANSI	Carbon Steel	3XWC, 3SDX
4" XP	A10-40B			4XWC, 4XD, 4SDX
4" XP	A10-60B	6" 150 lb. ANSI		4XWC, 4XD, 4SDX

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Appendix F

Electrical Calculations and Cutsheets

Appendix G

Fire Protection Calculations and Cutsheets